John W. Taylor
Department of Chemistry and Chemical Biology, Rutgers University, Piscataway, New Jersey 08854.

The Synthesis and Study of Side-Chain Lactam-Bridged Peptides

Abstract: *Side-chain lactam bridges linking amino acid residues that are spaced several residues apart in the linear sequence offer a convenient and flexible method for introducing conformational constraints into a peptide structure. The availability of a variety of selectively cleavable protecting groups for amines and carboxylic acids allows for several approaches to the synthesis of monocyclic, dicyclic, and bicyclic lactam-bridged peptides by solid-phase methods. Multicyclic structures are also accessible, but segment-condensation syntheses with solution-phase cyclizations are most likely to provide the best synthetic approach to these more complex constrained peptides. Lactam bridges linking (*i, i + *3)-, (*i, i + *4), and (*i, i + *7)-spaced residue pairs have all proven useful for stabilization of* α *helices, and (*i, i + *3)-linked residues have also been demonstrated to stabilize* β*-turns. These structures are finding an increasing number of applications in protein biology, including studies of protein folding, protein aggregation, peptide ligand–receptor recognition, and the development of more potent peptide therapeutics. Defining the functional roles of the amphiphilic* α*-helices in medium-sized peptide hormones, and studying helix propagation from rigid,* α*-helix initiating bicyclic peptides are among the most exciting developments currently underway in this field.* © 2002 Wiley Periodicals, Inc. Biopolymers (Pept Sci) 66: 49–75, 2002

Keywords: *lactam bridge; conformational constraints; peptide synthesis; helix stabilization; protecting groups; peptide structure; crosslinked peptides; protein folding; peptide hormones*

INTRODUCTION

The introduction of covalently bonded cyclic structures into polypeptides has long been used as an important approach to limiting the inherent flexibility of the peptide backbone and side chains.[1,2] Cyclization of peptides to create conformationally restricted analogs has been advantageous in a wide variety of fields involving peptide science. These include explorations of protein folding, determination of the biologically active conformations of peptide hormones and the related problem of designing peptidomimetics as potential therapeutic agents, and the development of new peptide-based materials. For example, end-to-end cyclization of short peptides has been used to generate self-assembling nanotubes[3] and antibacterial peptides[4]; the introduction of backbone-linked side chains [5] and side-chain to side-chain cross-links[6–8] has been used extensively to develop models for the receptor-bound conformations of the enkephalins and other small peptide hormones and neurotransmitters; and the introduction of cyclopropyl groups into side chains has found utility in generating local conformational restrictions in such bioactive peptides.[9]

The constraint of small synthetic peptides through the introduction of even one cyclic element can often

Correspondence to: John W. Taylor; email: taylor@rutchem.rutgers.edu

Contract grant sponser: NIH; contract grant number: R21 AI45316

Biopolymers (Peptide Science), Vol. 66, 49–75 (2002)

© 2002 Wiley Periodicals, Inc.

result in a highly constrained structure with well-defined limits on the conformations of the peptide backbone and the side chains.[7] In this case, the conformational entropy of the peptide may have been reduced by as much as 1–2 kcal/mol per constrained residue.[10] For bioactive peptides without such conformational constraints, much of this conformational entropy must be lost upon binding to their cognate receptors. Therefore, the introduction of such a constraint has the potential to generate very high affinity ligands with high selectivities for specific receptors among several homologous subtypes. In addition, other potentially advantageous characteristics may emerge, including increased resistance to proteolysis *in vivo*, enhanced uptake into the central nervous system, or controlled agonist or antagonist character at cell-surface receptors.

The introduction of cyclic structural elements into proteins has also been explored. Side-chain cross-linked proteins, and even backbone cyclized proteins, may be developed to have increased thermal stabilities, provided that these cross-links can be introduced without perterbation of the native, folded conformation.[11–14] As for small peptides, the benefits of such cyclizations in proteins are based on the entropic destabilization of the denatured form, since the goal is normally to retain the native conformation of the folded protein. The magnitude of this effect is dependent on the ring sizes introduced, their number, and their topologies.[11] In comparison to small peptides, the larger rings generally introduced into proteins by cross-linking side chains rarely limit the intervening peptide backbone to a single conformation. However, the introduction of multiple cross-links into overlapping positions in the peptide chain will generally be more effective as a conformational constraint than if those bridges were placed in completely separate segments of the primary structure. Numerous naturally occurring examples of the severe conformational constraints that can be imposed on the peptide backbone by such overlapping side chains are provided by peptide neurotoxins, such as the bee-venon peptides apamin or mast-cell degranulating peptide,[15,16] or the conotoxins.[17] These peptides incorporate multiple disulfide bridges placed in overlapping positions in their primary structures, and they are typically folded into a single, well-defined conformation, despite being much too small to form a globular structure with a hydrophobic core.

This article reviews some of the synthetic approaches that have been used to prepare peptides that incorporate side-chain to side-chain lactams bridges into their structures, as well as the conformational effects of these bridges and their applications to studies of bioactive peptides. The use of amide bond formation to cross-link side chains offers two significant advantages over the obvious alternative of using disulfide bridges. First, the multiple amide bonds in lactam-bridged peptides are chemically inert to most of the conditions to which peptides are subjected. In contrast, disulfide bonds at neutral pH may exchange to relieve strain or otherwise achieve a more stable folded conformation, rendering the conformational constraints imposed on a peptide by two or more disulfide bridges uncertain for most applications. Furthermore, disulfide-bridged peptides will be reduced to their acyclic thiol form in an intracellular environment, so that applications involving cellular uptake of such peptides are excluded. Second, the availability of a wide variety of protecting groups for amines and carboxylic acids that are cleavable under orthogonal or highly selective conditions makes the preparation of complex multicyclic peptides with overlapping bridge constraints more manageable than would be the synthesis of peptides incorporating multiple disulfide bridges. Typically, side-chain lactam bridges have been used to link amino acid residues that are separated in the peptide chain by at least two intervening, unconstrained residues, but not more than six. Therefore, the conformational constraints that these bridges impose on the peptide chain individually are usually intermediate in character between those of the small-ring constraints introduced into peptide neurotransmitters to generate highly defined local conformations, and those of the large-ring cross-links that have been introduced into proteins to generate a greater resistance to denaturation. However, just as multiple overlapping disulfide bridges impart a rigid backbone structure to the smallest peptide neurotoxins, the placement of even two lactam bridges in overlapping positions in a synthetic peptide structure can produce highly stabilized, unique conformations in intermediate-sized peptide structures.

SYNTHETIC APPROACHES

Compared to solid-phase peptide chemistry, the convergent approach of solution-phase methods allows a greater variety of ways to meet the requirements for selective protection and deprotection of the backbone and side-chain functional groups that are essential to the synthesis of lactam-bridged peptides. In addition, solution-phase syntheses of cyclic peptides allow direct monitoring of the progress of each reaction step, the purification of key intermediates as necessary, and a high level of control over the reaction conditions. In particular, the concentration of the peptide during the

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

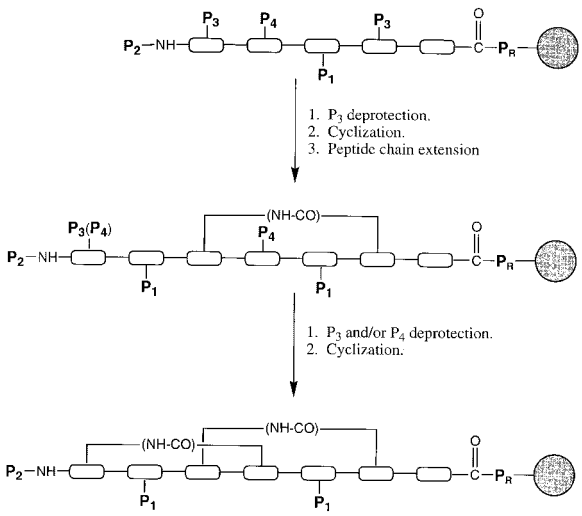


FIGURE 1 Generalized scheme showing the protecting group requirements for the preparation of monocyclic and bicyclic side-chain lactam-bridged peptides by solid-phase synthesis. Amino acid residues (open shapes) and the polymer support (shaded circle) are indicated schematically with their associated protecting groups (P_n).

critical cyclization reaction(s) can be adjusted to any level necessary to maximize the yield by minimizing interchain reactions.[18] However, most peptide chemists have preferred to adapt and develop the more expedient solid-phase approach to prepare such cyclized peptides for research applications. The solid-phase approaches to lactam-bridged peptides will be discussed here, but their extension and adaptation to include solution-phase cyclizations, and allow the preparation of multicyclic structures with overlapping lactam bridges will also be introduced.

The requirements for solid-phase assembly of side-chain cyclized peptides are summarized in schematic terms in Figure 1. In this scheme, the linkage chemistry to the solid support (carboxyl derivative P_R) is necessarily considered to be part of the protection strategy, and the nature of this linkage structure will depend on whether a C-terminal carboxylic acid or amide is ultimately desired. As with any solid-phase synthesis, side-chain functional groups require semi-permanent protection with groups (P_1) that will normally be selected to have the same stability profile as the linkage to the solid support. A rapidly cleaveable temporary protecting group, P_2, is also required for the α-amine of each amino acid derivative during its coupling reaction. In the *t*-butoxycarbonyl (Boc)/benzyl strategy for solid-phase synthesis,[19,20] the P_R and P_1 groups are of the benzyl type, and are usually removed during peptide cleavage from the solid support by liquid HF or other strongly acidic conditions.[21–24] P_2 is the Boc group, which is cleaved in each cycle by 50% triflouroacetic acid (TFA) in dicloromethane (DCM). In contrast, the 9-fluorenyl methyoxycarbonyl (Fmoc) method[25] uses the Fmoc protecting group for P_2, with the advantage that it is truly orthogonal to the acidolytic methods used to cleave the P_R and P_1 groups. In Fmoc syntheses, the P_1 groups are usually of the *tert*-butyl type, and cleavable by concentrated TFA. The linkage to the solid support, P_R, is also designed for cleavage by TFA, and benzyl groups that are highly activated for acidolytic cleavage are usually used. The standard Fmoc method therefore sacrifices one level of differentiation in the lability of the protecting groups toward acidolysis, but the orthogonality of this protection scheme and the milder acid conditions employed for the final step give the advantage that cleaner products are usually observed. Furthermore, the P_R and P_1 groups may also be of the standard benzyl type used for Boc chemistry, if an extra level of selectivity during acidolytic protecting group removal is required.

The preparation of a monocyclic lactam-bridged peptide by solid-phase methods will generally require a pair of selectively cleavable protecting groups of type P_3 used for protection of the amine and carboxylic acid to be linked by solid-phase cyclization (Figure 1). In the first report of the synthesis of a monocyclic lactam-bridged peptide by cyclization on the solid support, the Schiller group reported the use of Fmoc chemistry (P_2 = Fmoc) on Merrifield resin (P_R = benzyl type group cleaved by strong acid) to prepare (Lys, Glu), (Orn, Glu), and (Glu, Lys) side-chain-linked enkephalin analogs, with the lactam-bridged side chains placed in the second or third and fifth residue positions.[26,27] Since these peptides did not require orthogonal side-chain protection in the unbridged positions (P_1 groups), selective cleavage of the bridging groups was achieved by using TFA to cleave *tert*-butyl type protecting groups (Boc and *tert*-butyl ester) that are standard for the Fmoc method. The cyclization reaction was carried out over 5–10 days, using the dicyclohexyl-carbodiimide/1-hydroxybenzotriazole (DCC/HOBt) active-ester method, and the monocyclic and cyclodimerized peptides were the major components identified in the crude peptide mixtures by high performance liquid chromatography (HPLC)

In 1988, Felix and co-workers reported improved yields in the synthesis of side-chain lactam bridged GRF analogs by Boc/benzyl chemistry, when the benzotriazole-l-yl-oxy-tris(dimethylamino)-phosphonium hexaflourophosphate (BOP) coupling reagent was used with HOBt in place of DCC/HOBt for the cyclizations. In these syntheses, side-chain protecting

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

groups at a third level of selectivity, P_3, *were* required to direct the specific location of the side-chain bridging cyclization. The base-labile Fmoc-amine and OFm-ester protecting groups were used for this purpose. Then, by simply performing the side-chain Fmoc/OFm deprotections and cyclizations at separate stages in the peptide chain assembly, this strategy also allowed the synthesis of dicyclic peptides, in which two side-chain lactam bridges were incorporated into GRF analogs in nonoverlapping positions in the peptide chain.[28,29]

The Boc/benzyl/Fm protection strategy has been used by numerous groups since the initial report by Felix et al.,[28] but the BOP reagent has generally been replaced by PyBOP or other modern coupling reagents of the phosphonium-ion type, to avoid generating carcinogenic byproducts. Uronium-ion coupling reagents such as 2-(1H-benzotriazole-l-yl)-tetramethyluronium hexaflourophosphate (HBTU) or TBTU have also been used successfully as the coupling reagent for cyclizations. However, if the desired side-chain cyclization reaction is slow, competing side reactions can become dominant in which the uronium carbon is transferred onto the amine, irreversibly generating tetra-alkylated guanidinum groups and blocking the cyclization reaction.[30] These reagents, therefore, are now generally avoided.

Slow cyclization to form the desired *intra*chain cyclization will often result in a poor or failed synthesis of a lactam-cyclized peptide, for the simple reason that *inter*chain amide-bond formation on the solid support will become competitive. Cyclodimerized and polymerized peptides then become the dominant products. In order to minimize this problem, it is worthwhile optimizing the conditions for a side-chain cyclization during solid-phase synthesis, by performing the reaction on a small scale and testing various solvents or solvent mixtures, different coupling reagents, and different amounts of base [usually diisapropylethylamine (DIEA)]. Complete deprotection and cleavage, followed by analysis of the products by HPLC and mass spectrometry, allows an assessment of the optimal conditions. Typically, polymeric products appear on the HPLC profile as a broad "hump" in the baseline absorbance, over which a sharp absorbance peak due to the desired cyclized product is superimposed. Since HPLC is the only high-resolution purification method normally used for the separation of the correct cyclic product from all of these other byproducts, the presence of polymerized peptides as a large fraction of the total peptide absorbance in an HPLC elution profile greatly undermines the confidence a researcher may have in the purity of the isolated major product.

Through the optimization of a large number of solid-phase lactam-bridge cyclizations during peptide synthesis in my laboratory, a number of general observations have been made. First, one of the major determinants of the cyclization yield is the amino acid sequence context (the "host" peptide) of the residues to be bridged. It is surprising, perhaps, that this sequence effect can be even stronger than the effect of altering the ring size of the cyclization product. For example, we have successfully linked amine and carboxylic acid groups on side chains separated by three or six intervening residues, creating macrocyclic structures ranging in size from 20-atom rings to 34-atom rings, and have obtained monocyclic or even dicyclic peptide products in good yields in both types of reaction.[31,32] However, monocyclic analogs of human calcitonin in which the side chains of aspartic acid and lysine residues in positions 17 and 21 were linked to form a 20-atom macrocycle have consistently given much lower yields.[33,34] Similar effects of the sequence context have been reported by Hearn and co-workers.[35] These authors have also reported that the presence of an Fmoc-protected amino acid residue immediately N-terminal to the residues being cyclized improved the cyclization result, presumably by limiting the conformational freedom of the peptide chain during the reaction.

Another general observation that we have made concerning the on-resin cyclization reactions is that lowering the peptide-chain substitution levels by about a factor of two, from the typical level of 0.30–0.40 meq/g for the C-terminal amino acid residue down to a level of 0.15–0.20 meq/g, has little effect on the purity or yield of the desired cyclic peptide in the crude products. Therefore, this approach of diluting the peptide chains on the solid support is not usually worthwhile, since it is expensive in terms of reagents and solvents. A related result of these optimization processes is that we have typically found that increasing the polarity and hydrogen-bonding character of the solvent used during the cyclization reactions can often lower the polymerization levels and improve yields considerably. For example, DCM is usually a poor choice of solvent for on-resin cyclizations, and dimethylformamide (DMF) and N-methylpyrolidone (NMP) are good solvents. In one study where cyclodimerization was a problem, we found that a DMSO/NMP (1/4) mixture gave excellent results, improving the monomer:dimer product ratio in one case from zero in DCM/DMF (1/1) up to 65:35 in the optimized solvent mixture, and from 40:60 up to 95:5 in another case.[36] These examples appear to indicate that aggregation of the peptide chains on the solid support is an important factor contributing to polymerization, and that effective solvation of the peptide backbone can greatly reduce this

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

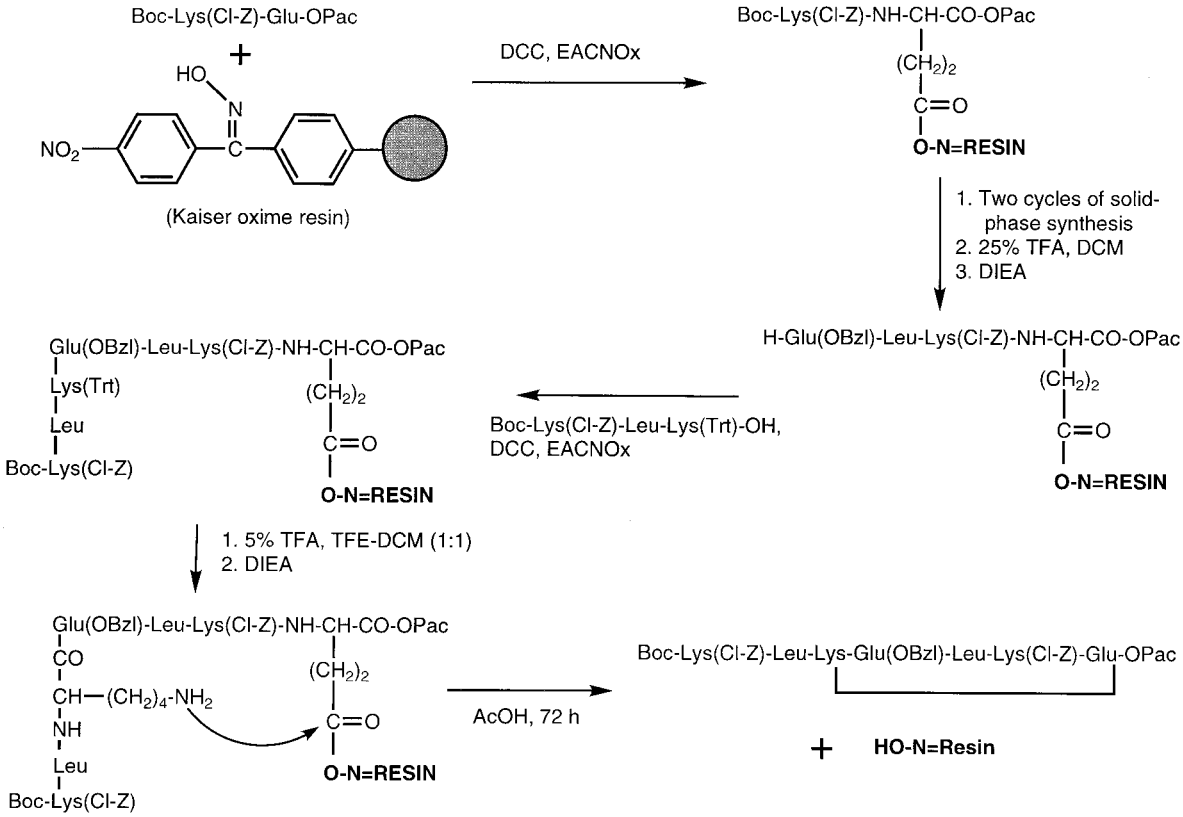


FIGURE 2 Synthesis of a fully protected monocyclic lactam-bridged peptide segment using oxime resin. The peptide chain is assembled on the Kaiser oxime resin, using a side-chain oxime-ester linkage to the solid support. Selective deprotection of a side-chain amine allows side-chain cyclization to form the lactam bridge with concomitant cleavage from the resin. The crude monocyclic peptide product is obtained in solution, and is typically free of polymeric products, rendering purification unnecessary. The N- and C-terminal protecting groups allow for the solution-phase segment-condensation assembly of multicyclic peptides.[37,38]

problem. Finally, it is worth noting that the basicity of the cyclization reaction may also be important, and can be difficult to control, particularly if the side-chain protecting groups have just been cleaved by acid, since residual reagents may be present at unknown levels after washing the resin. To achieve consistency in the cyclization conditions, we have found that the addition of base (DIEA) to the coupling reaction until a pH of about 8 is indicated on wet litmus paper is effective.

The interchain reactions that result in polymeric by-products during solid-phase cyclization reactions can often result in dramatically low yields when a second cyclization reaction is performed during a single peptide chain assembly. This makes the synthesis of dicyclic lactam-bridged peptides by solid-phase methods significantly less reliable than that of monocyclic peptides, and attempts to perform three successive cyclizations on the solid support have rarely been reported. In order to address this problem, and wishing to combine the advantages of rapid peptide chain assembly on solid supports with the advantages of convergent syntheses by solution-phase methods, we have proposed the synthesis of protected, side-chain cyclized peptide building blocks by using the Kaiser oxime resin.[37,38] In this approach (Figure 2), the P_R linkage is a *p*-nitrobenzoyl oxime ester and this group is used to link the side-chain carboxylic acid of one bridging amino acid to the resin. The peptide chain is built up using Boc/benzyl protection chemistry (P_1 and P_2). Then cyclization with concomitant cleavage of the protected monocyclic product is achieved by selective deprotection of the amine component of the bridge, which is then allowed to react with the oxime ester linkage. The monocyclic products obtained by this method are usually found to have very high purity, since polymeric byproducts tend to remain attached to the solid support. The unpurified protected segments can then be deprotected at their N- or C-termini and linked to each other to produce

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

multicyclic products. Alternatively, the C-terminal deprotected product could be coupled to a peptide chain that is assembled on a solid support by the usual Boc/benzyl methods.[39] A key protecting group in this synthetic approach is the trityl group on the side chain amine, which provides a third level of selectivity to the acidolytic cleavages. Selectivity in the cleavage of the C-terminal carboxylic acid was provided by protection as the phenacyl ester (OPac), which is stable to acidolysis and can be reductively cleaved under selective conditions (Zn/AcOH). This group was later replaced by the allyl ester (OA1), which is stable to both acid and base, and gave cleaner deprotection reactions in solution, using the catalyst tetrakistriphenylphosphine Pd^0 in the presence of a buffered secondary amine.[40]

Use of allyl ester protection to develop side-chain lactam bridge cyclizations was first introduced by Felix and co-workers in their studies of GRF.[29,41] In these studies OA1 protection was used for the side chain of Asp^{25} in syntheses by the Boc/benzyl method. Continued solid-phase assembly of the peptide chain after $Asp(OA1)^{23}$ incorporation to include Fmoc protection of the side-chain amine of Orn^{21}, then allowed for selective deprotection of the Orn^{21} side chain with base, so that Fmoc–Gly–OH could be coupled to this amine, and used as a spacer in a lactam bridge joining Orn^{21} to Asp^{25}. Deprotection of the α-amine of the Orn^{21} side-chain-linked Fmoc–Gly residue, and cleavage of the OA1 side-chain-protecting group on Asp^{25}, then allowed the macrocyclization on the resin to be performed, generating an $Orn(Gly)^{21}$-to-Asp^{25} side-chain lactam bridge.

Introduction of the OA1-protecting group for side-chain carboxylic acids, along with its urethane-based partner for side-chain amines, the Aloc group, and the demonstration that these groups can be reliably cleaved in good yield on solid supports using the Pd^0 catalyst, provided another level of orthogonality in side-chain protection. This has opened up the possibility of using the allyl-protecting groups as the P_3 type (Figure 1) needed for solid-phase assembly of lactam-bridged peptides by standard Fmoc/*tert*-butyl methods. The Fmoc/*tert*-butyl/Al combination has been demonstrated to be very effective in generating high-purity products,[18] and has recently been rendered more reliable by the use of Giubé's method for allyl group cleavage by Pd^0-catalyzed hydrostannolysis.[42–44] Furthermore, the Aloc/OA1-protecting group pair may be used in Boc/benzyl syntheses as protecting groups P_4, in combination with the Fmoc/OFm groups as P_3 (Figure 1), allowing the synthesis of highly constrained peptides with overlapping lactam bridges incorporated into their structures in either bridge orientation. This synthetic scheme is illustrated with the specific example shown in Figure 3A.[45]

As an alternative to the Boc/benzyl/Fm/Al scheme, we have also explored the use of 2-(2-pyridyl)ethyl esters (OPet). OPet esters are rendered base-labile by methylation of the pyridyl nitrogen,[46,47] but the original methylation conditions must be modified to produce high yields on the solid support.[48] Cleavage of the methylated side-chain-protecting group is then achieved under conditions essentially identical to those used to remove the Fmoc and OFm groups (morpholine/DMF, 3/7), by the same β-elimination mechanism. Since amine protection by the corresponding urethane group (2-[2-pyridyl]ethyloxycarbonyl) is also possible, these 2-[2-pyridyl]ethyl-based (Pet)-protecting groups may also be combined with the Boc/benzyl/Fm protection scheme. This again allows the preparation of bicyclic peptides with lactam bridges having either directionality, using only solid-phase cyclizations. The original demonstration of this method is illustrated in Figure 3B.[48]

Most recently, we have preferred to extend the Fmoc method for peptide chain assembly to include a second level of acidolytic selectivity, in order to prepare monocyclic side-chain lactam-bridged peptides.[34] In doing so, we have chosen to avoid the use of benzyl-type protecting groups for the P_1 level of protection, in order to avoid the need for strong acidolytic cleavage conditions (HF, TFMSA, etc.). Instead, we have sought to exploit the increased acid lability of highly activated benzyl groups, and to use these at the P_3 level in combination with the *tert*-butyl type groups at the P_1 level. This approach seemed reasonable, based on our earlier experience with side-chain cleavage of the trityl group on lysine in the presence of N^{α}-Boc protection, during the course of the oxime-resin syntheses described above (Figure 2). The 4-methyltrityl (Mtt) side-chain protected derivatives of Fmoc–Dap–OH, Fmoc–Dab–OH, Fmoc–Orn–OH, and Fmoc–Lys–OH, as well as the phenylisopropyl ester (OPip) side-chain protected derivatives of Fmoc–Asp–OH and Fmoc–Glu–OH, are all now commercially available. This allows the synthesis of a wide variety of monocyclic lactam-bridged peptide analogs by combining Fmoc/*tert*-butyl chemistry with the selective cleavage of Mtt and OPip protecting groups with 1% TFA in DCM, prior to specific cyclization of the deprotected amine and carboxylic acid groups on the solid support. Mtt and OPip deprotections may be followed by monitoring the release from the solid support of these uv absorbing groups, so that loss of other side-chain-protecting groups of the *tert*-butyl type is minimized by keeping the deprotection reacton short (1–2 h). In our experience, this depro-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

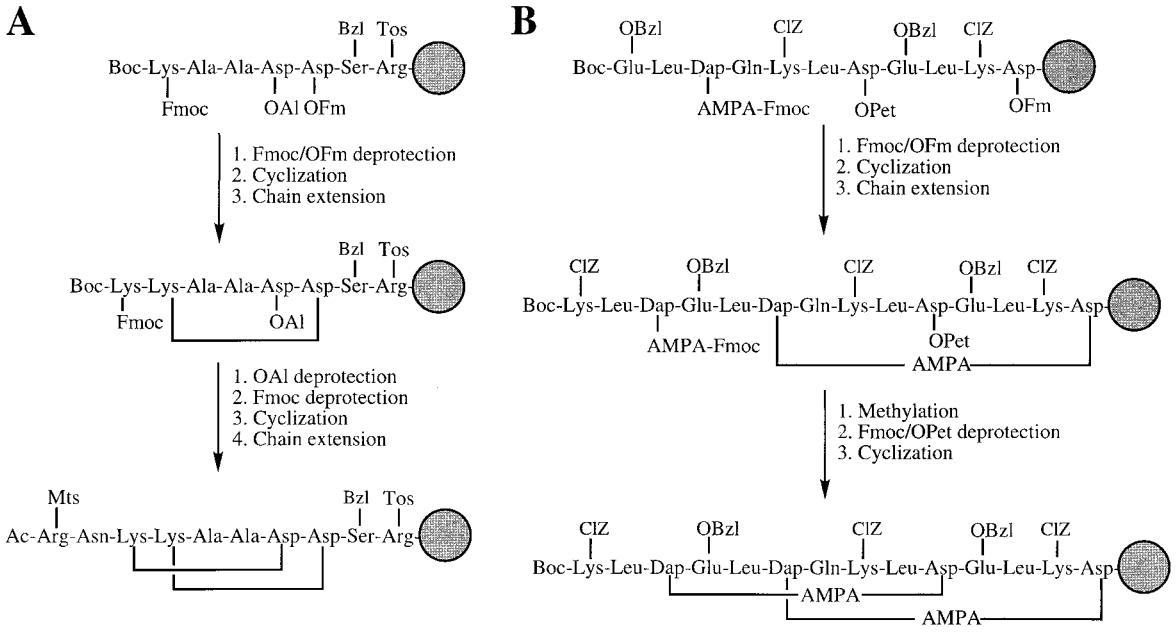


FIGURE 3 Synthesis of bicyclic lactam-bridged peptides using solid-phase methods. Two synthetic approaches that employ Boc/benzyl chemistry for the peptide chain assembly are illustrated. Side-chain protecting groups Fmoc/OFm and Aloc/OAl are employed in the synthesis of a model peptide designed to study helix propagation from a bicyclic (Lys^{i}, Asp^{i+4})-bridged hexapeptide[45] (A); side-chain Fmoc/OFm protection is combined with the masked base-lability of the OPet-protecting group in the synthesis of a bicyclic model peptide designed to study helix stabilization by the (i, i + 7) AMPA bridge[48] (B).

tection method is more easily reproducible than Pd^{0}-catalyzed Aloc/OA1 deprotection, which can fail if the catalyst is improperly stored and becomes deactivated, or if an inert atmosphere is not rigorously maintained during the reaction. And in comparison to the use of Pet protection, Mtt/OPip protection has the advantage of avoiding the extra methylation step, which cannot be monitored directly for completion. One minor disadvantage of Mtt protection for the amine component of the bridge is that, as long as the Mtt-protecting group is present, the Kaiser ninydrin test for completion of coupling reactions during peptide chain assembly[49] is unreliable, as it gives false positives. This problem can usually be overcome, however, if the amount of the KCN/pyridine reagent used in the test is doubled.[34]

As described above, at least three reliable protecting group strategies are now available for the synthesis of bicyclic peptides through the exclusive use of solid-phase methods (Boc/benzyl/Fm/A1; Fmoc/*tert*-butyl/Pet/OPip or Mtt; or Fmoc/*tert*-butyl/A1/OPip or Mtt), and other combinations may also be considered. However, the difficulties of obtaining good yields with minimal polymeric products from solid-phase syntheses remain, and it seems unreasonable to attempt to perform more than two cyclizations on the solid support during any single peptide chain assembly. Nevertheless, we would indeed like to be able to synthesize peptides that have three or more lactam (or other) cyclic constraints incorporated into their structures. These goals might best be achieved, therefore, by combining the efficiency of solid-phase peptide chain assembly with solution-phase cyclizations to generate cyclized building blocks for subsequent segment-condensation assembly into more complex structures. The Rink acid resin, which was designed for the preparation of protected peptide segments in solution by using solid-phase Fmoc/*tert*-butyl methods,[50] appeared to us to have potential application to this problem. The trialkoxydiphenylmethyl ester linkage of the peptide C-terminus to the Rink acid resin is very sensitive to acidolysis, and is cleaved by 10% acetic acid in DCM. Due to this extreme acid lability, coupling reactions on the Rink acid resin must not be performed over extended times, and are best performed under buffered conditions, with one equivalent of base (DIEA) added for each equivalent of Fmoc-amino acid *and* HOBt (or equivalent reagent) used in the coupling reaction. Despite these potential problems, good yields of short protected peptides are readily obtained. Recently, we reported that peptide cleavage from the Rink acid resin may be performed

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

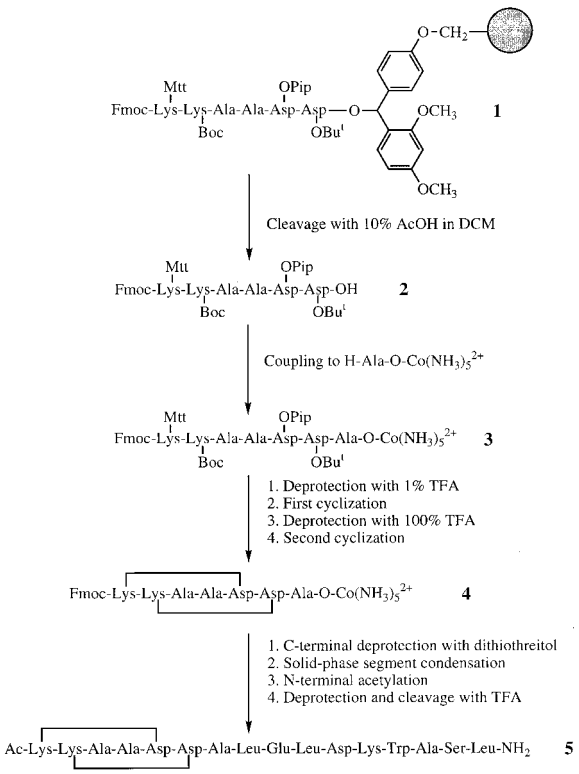


FIGURE 4 Synthesis of a bicyclic lactam-bridged peptide using solid-phase peptide chain assemblies, solution-phase cyclizations and segment condensation. The peptide link to the Rink acid resin provides a fourth level of acid lability, and allows the synthesis of protected peptide **2** by Fmoc/*tert*-butyl methods incorporating the acid-labile Mtt and OPip groups. Two separate solution-phase cyclizations may then be specified to yield the protected bicyclic peptide **4**. Peptide **4** is suitable for segment condensation syntheses in either direction. This is illustrated by the solid-phase assembly of peptide **5,** which was designed to study helix propagation in the membrane-proximal region of the HIV-1 surface glycoprotein, gp41.[51,54]

without loss of either the Mtt group on the ε-amine of Lys, or the OPip ester group on the β-carboxyl of the Asp side chain.[51] Based on this discovery, we have synthesized the protected peptide Fmoc–Lys(Mtt)–Lys(Boc)–Ala–Ala–Asp(OPip)–Asp(Boc)–OH (**2**) using the Rink acid resin. In a typical synthesis, 700 mg of the crude protected peptide **2** was obtained in good purity, starting from 1.0 g of the Rink resin. As illustrated in Figure 4, this *fourth* level of selectivity just in acidolytic cleavage of protecting groups (benzyl < *tert*-butyl < Mtt/OPip < Rink acid resin) has allowed us to synthesize the bicyclic peptide building block Fmoc-(*cyclo*-1-5, 2-6)-[Lys–Lys–Ala–Ala–Asp–Asp–Ala]–O–$Co(NH_3)_5^{2+}$ (**4**), using solution-phase cyclizations.

In the synthesis described in Figure 4, the temporary introduction of another selectively cleavable carboxyl-protecting group was also required, to allow specific control of the cyclization reactions. For this additional level of carboxyl protection, we chose to use the exchange-inert cobalt(III) pentaammine complex.[52] This group is stable to both acid and base, and is quantitatively cleaved by mild reduction to the exchange-labile cobalt(II) form in aqueous solution, using (for example) dithiothreitol. Furthermore, the presence of a formal +2 charge on the cobalt(III) pentaammine carboxylate complex is also extremely advantageous. First, it endows the protected peptide intermediates in the reaction scheme shown in Figure 4 with enhanced solubilities both in the organic solvents, such as DMF, that are used for all of the amide-bond forming reactions, and in the mixed organic-aqueous solutions (CH_3CN–water, or DMF–water) that are used for chromatography. Second, the presence of a cationic charge allows ion-exchange chromatography to be used. In combination with reversed-phase chromatography, highly effective purifications were achieved during the preparation of **4**, using simple low-pressure, low-resolution, step-gradient approaches.[51]

The potential utility of lactam-bridged protected peptide segments such as **4** has been demonstrated through the preparation of a helix-initiated bicyclic lactam-bridged peptide from the HIV envelope glycoprotein, gp41, using a solid-phase segment-condensation approach (Figure 4).[53,54] With efficient and scaleable methods in place for the synthesis of the necessary cyclized intermediates, it should now be possible to extend the synthetic approach described in Figure 4 to allow the preparation of an infinite variety of multiply cyclized lactam-bridged peptides. In principle, any sequence may be accommodated, even within the lactam-bridged segments, because benzyl-type protection can be employed for the unbridged side chains in the cyclized segments. Many different multicyclic topologies are then accessible, through the repeated segment-condensation of individual monocyclic and bicyclic protected peptide building blocks.

CONFORMATIONAL CONTROL OF PEPTIDES USING SIDE-CHAIN LACTAM BRIDGES

Stabilization of the α-Helical Conformation Using (*i*, *i* + 4) Side-Chain lactam Bridges

The most commonly reported use of side-chain lactam cyclizations in peptides has been for stabilization of

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the α-helical conformation. For this purpose, peptide analogs with lactam bridges linking side-chain pairs in the *i*th and $(i + 4)$th position of the peptide chain, which are in close proximity on the surface of an α-helix, have usually been used. Helix stabilization *via* complex formation of $(i, i + 4)$-spaced His side chains to metal ions[55–57] or to palladium(0),[58] and by hydrazone bonds substituted into the peptide backbone in place of the helical CO^{i}-to-NH^{i+4} backbone H-bond[59] have also been described, but will not be reviewed here.

The first determination of a helix stabilizing effect for side-chain lactam bridges came from Felix and co-workers, who incorporated Asp^{i}, Lys^{i+4} side-chain linkages, and the reverse Lys^{i}, Asp^{i+4} side-chain linkages, into GRF-(1–29) analogs to link residues 8 and 12 and residues 21 and 25, respectively.[28,41] Studies of a linear, unbridged GRF analog in water at pH 3, by CD and NMR spectroscopy, demonstrated that peptide segments in the N-terminal region at residues 7–14 and in the C-terminal region at residues 21–28 naturally favored the helical conformation. When placed within these peptide sequences, both the Asp-to-Lys and the Lys-to-Asp linkages increased the fractional helix content of these GRF analogs without causing any significant distortion of the helical structures. Also, the effects of the individual bridges on the overall helix content were approximately additive in the dicyclic peptide.[28,29,60]

Later studies from the Felix group and others[29,38,61–63] have investigated the helix-stabilizing effects of alternative lactam ring sizes and bridge orientation in *i*-to-$(i + 4)$ side-chain links. For helix stabilization, the 20-atom ring formed by an (Asp^{i}, Lys^{i+4}) or a (Lys^{i}, Asp^{i+4}) bridge is optimal. Smaller ring sizes (19 atoms or less) are clearly helix destabilizing.[29,63] Molecular modeling studies indicate that this result is a direct consequence of these bridges being too short to connect across one turn of α-helix. Adding one methylene group to the ring by using a Lys-to-Glu side-chain linkage (21 atoms) has been reported to give lesser helix stabilizing effects[38] or no helix stabilization at all,[61] although 21-atom bridges having the reverse orientation (Glu^{i}, Lys^{i+4}) appear to have a strong stabilizing effect.[61] Interestingly, certain lactam-bridged GRF analogs with larger ring sizes of up to 24 atoms also appear to have a helix-stabilizing effect, based on their reported CD spectra.[29] These bridges, which were generated through the introduction of "spacer" amino acid residues such as Gly, β-Ala, and γ-Aba into the side-chain bridge, must retain considerable flexibility in the helical conformation. In contrast, NMR studies of bicyclic Lys^{i}, Asp^{i+4}-bridged peptides[45,64] indicate that these bridges adopt well-defined conformations on the helix surface, with the bridging amide nitrogen donating a hydrogen bond to the backbone Lys^{i} carbonyl group.

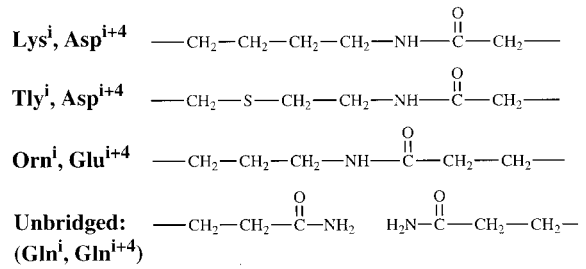


FIGURE 5 Structures of $(i, i + 4)$ bridges studied for helix stabilization. The $(i, i + 4)$ side-chain bridges (Tly^{i}, Asp^{i+4}) and (Orn^{i}, Glu^{i+4}), or unbridged glutamine residues, were incoporated into the model peptide illustrated in Figure 6A, in place of the two (Lys^{i}, Asp^{i+4}) bridges, in order to compare the helix stabilizing effects of these structures (Figure 7).[62]

In an attempt to optimize the helix stabilization that can be achieved through the introduction of a 20-atom, macrocyclic, *i*-to-$(i + 4)$ side-chain link, we developed a dicyclic 14-residue model peptide system for comparing the helix-stabilizing effects of various lactam bridges by CD spectropolarimetry (Figures 5 and 6).[62] Our point of reference was the dicyclic (Lys^{3}, Asp^{7})- and (Lys^{10}, Asp^{14})-bridged peptide H-*cyclo*(3–7, 10–14)–[Lys–Leu–Lys–Glu–Leu–Lys–Asp]$_2$–OH, which has about 40% helical in aqueous solution at 25°C and pH 7.0. The helix content of this peptide was compared to that of an acyclic analog in which the bridged positions were substituted by glutamine residues, and to the helix contents of two dicyclic analogs. In one dicyclic analog, the β-methylenes of the bridging lysine residues were replaced by sulfur to give (Tly, Asp) bridges; in the other analog, (Orn, Glu) bridges were used, so that the bridging amides were moved towards the N-terminal end of the bridge by one methylene group (Figure 5). A comparison of the CD spectra obtained for each of these peptides is shown in Figure 7. These spectra demonstrate that helix stabilization in this 14-residue peptide is dependent on lactam bridge stabilization, since the linear analog exhibited very little helical structure (estimated as 5%). Interestingly, the (Orn, Glu) bridges actually appeared to have a helix-destabilizing effect, compared to the unbridged analog. Even the very conservative substitution of thiolysine into the bridges in place of the lysines results in a lower helix content for this peptide compared to the (Lys, Asp)-bridged analog. Overall, therefore, these results suggest that the helix stabilizing effects of the (Lys^{i}, Asp^{i+4}) bridge result from its having a very specific fit to the helix surface.

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

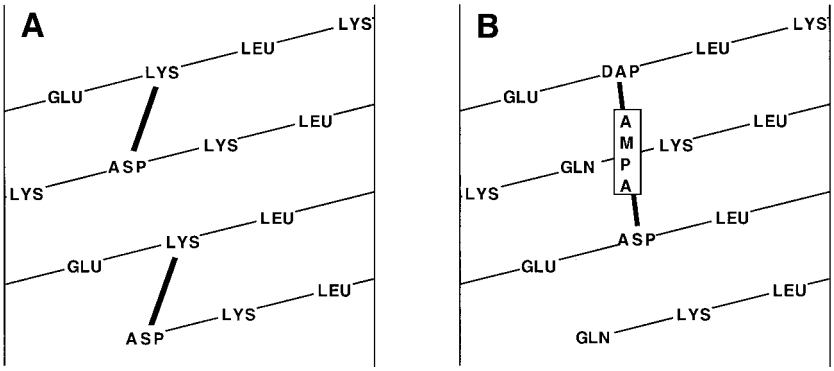


FIGURE 6 Helical net diagrams illustrating the structures of model peptides designed to study helix stabilization by side-chain lactam bridges. A model peptide H–(Lys–Leu–Xxx–Glu–Leu–Lys–Yyy)$_2$–OH was designed to study the helix stabilizing effects of substituting different (i, i + 4) bridges (Figure 5) into positions Xxx and Yyy. The (Lys^i, Asp^{i+4})-bridged dicyclic peptide is shown[62] (A). Adaptation of this model system to study (i, i + 7) bridges is illustrated using the monocyclic, (Dap[AMPA]^i, Asp^{i+7})-bridged peptide[32] (B). Side-chain bridge connections are shown as thick bars.

Assuming a two-state, helix–coil transition as a first approximation of the conformational equilibrations of these dicyclic 14-residue peptides, a simple analysis of the helix contents estimated from the spectra in Figure 7 suggests that the helical conformation of the (Lys, Asp)-bridged peptide H-*cyclo*(3–7, 10–14)–[Lys–Leu–Lys–Glu–Leu–Lys–Asp]$_2$–OH is favored over disordered structures by about 0.7 kcal/mol per lactam bridge, compared to the acyclic peptide H-(Lys–Leu–Gln–Glu–Leu–Lys–Gln)$_2$–OH.[62] This

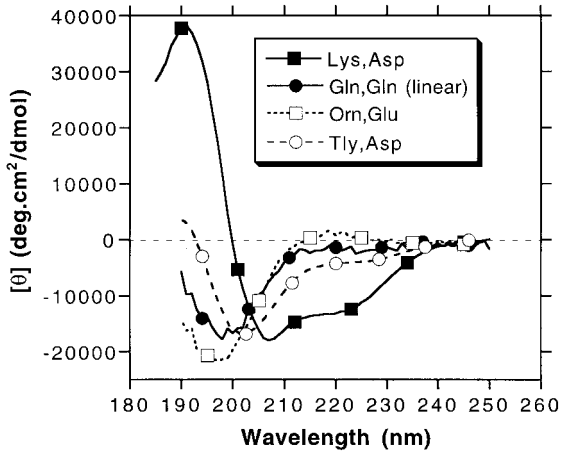


FIGURE 7 CD spectra of dicyclic (i, i + 4) lactam-bridged peptides. The CD spectra of the dicyclic peptides illustrated in Figures 5 and 6 were measured in aqueous phosphate buffer, pH 7.0, at 25°C. The fractional helix content, indicated by a maximum near 190 nm, and minima near 208 and 222 nm, was greatest for the (Lys^i, Asp^{i+4})-bridged peptide, and least for the (Orn^i, Glu^{i+4})-bridged peptide.

difference in helix stabilization is in the same range as those observed for different single-residue substitutions in host–guest systems that measure the helix–coil equilibrium.[65] Thus, it is not surprising to find that the local amino- acid sequence has a significant influence on the degree of helix stabilization that results from the introduction of any side-chain lactam bridge. For example, the amino acid sequence in residues 8–22 of salmon calcitonin (sCT), which corresponds to the potential amphiphilic α-helical structure,[66] is known to favor the α-helical conformation more strongly than does the corresponding sequence in human calcitonin (hCT). We have introduced an (Asp^{17}, Lys^{21}) side-chain lactam bridge into each of these peptides, replacing glutamine and threonine residues in the salmon peptide and asparagine and threonine residues in the human peptide.[34] The result of bridge substitution in sCT was a small increase in helix content in aqueous solution, as expected. In 50% triflouroethanol (TFE), it is known that sCT residues 8–22 are fully α-helical, and the helix content of the bridged sCT analog was identical to that of sCT. In the hCT sequence, however, the same (Asp^{17}, Lys^{21}) bridge substitution had a clear helix destabilizing effect in both solvent systems (Figure 8).

In another example,[67] substitution of the side-chain cyclized pentapeptide sequence *cyclo*(1–5)–[Lys–Ala–Ala–Ala–Asp] into a position at the N-terminal end of the DNA-binding basic region of the yeast transcription factor, GCN4, had a markedly smaller helix stabilizing effect on this peptide than did the same substitution at the C-terminal end of the same peptide (Figure 9). This difference most likely results

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

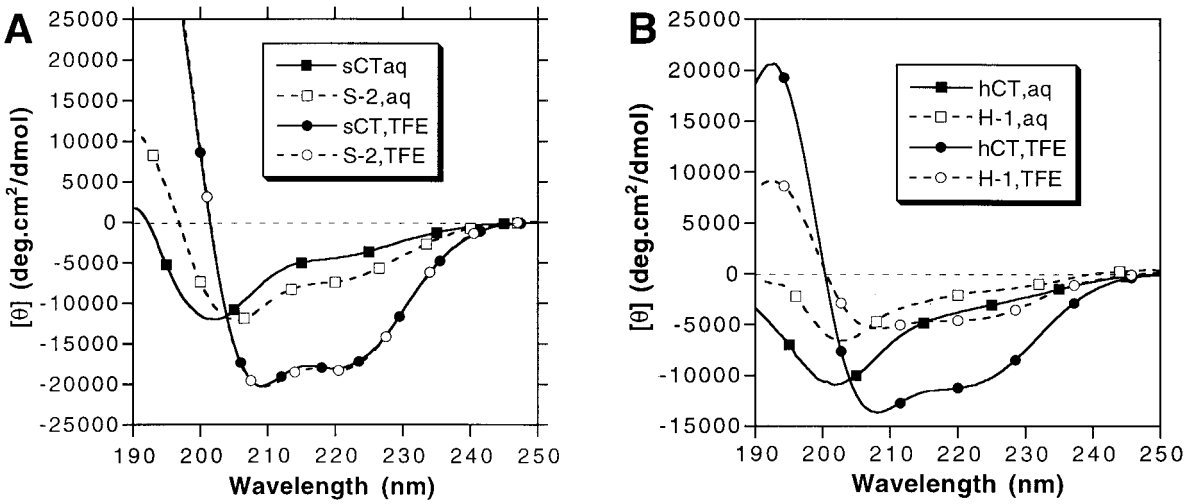


FIGURE 8 CD spectra of (Asp^{17}, Lys^{21})-bridged calcitonin analogs. The helix stabilizing effects of an (Asp^{17}, Lys^{21}) side-chain bridge incorporated into [His^{19}]-salmon calcitonin (peptide S-2) or into human calcitonin (peptide H-1) are compared. Peptide S-2 is more helical than sCT in aqueous buffer, pH 7.4, and both salmon peptides are highly helical in 50% TFE (A). Peptide H-1 is significantly *less* helical than hCT in aqueous buffer, pH 7.4, and in 50% TFE (B). Spectra were measured at 25 °C.[34]

from the fact that the substituted sequence at the N-terminal end of this peptide is already endowed with significant helix-stabilizing properties, due to the presence of the N-terminal helix capping residues Asp^1 and Pro^2 as well as two helix-stabilizing alanine residues. In contrast, the C-terminal terminal end of the peptides studied has no discernable helix-favoring characteristics.

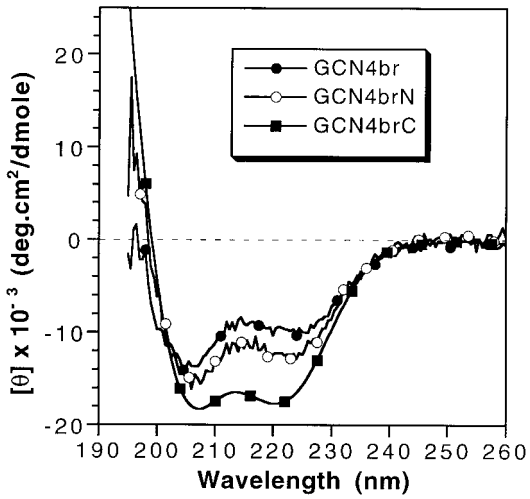


FIGURE 9 Different helix stabilizing effects of (Lys^i, Asp^{i+4}) bridges placed at the N-terminus and the C-terminus of the basic-region peptide from GCN4. The CD spectra measured in aqueous phosphate-buffered saline, pH 7.4, at 25°C, are shown for the unmodified basic-region peptide, GCN4br, the peptide bridged at the N-terminal end, GCN4brN, and the peptide bridged at the C-terminal end, GCN4brC.[67]

DNA binding by the basic-region peptide from GCN4, GCN4br, requires it to adopt a fully α-helical conformation, whereas it is predominantly disordered in its unbound form in aqueous solution. The lactam bridges in the N- and C-terminally helix-stabilized GCN4br peptides are located beyond the peptide–DNA interface in the complex and cannot have any direct interactions with the DNA. Therefore, differences in the stabilities of the peptide–DNA complexes formed by these lactam-bridged peptides compared to that of the linear peptide–DNA complex may be used as a measure of the helix stabilization introduced into the linear peptide through substitution of the lactam-bridged pentapeptide structure. A binding assay has been developed that measures DNA binding by these basic-region peptides by competitive displacement of a fluorescent-labeled peptide.[68] Using this assay, the more helical, C-terminally bridged peptide has been determined to bind to its cognate DNA sequence with a dissociation constant, K_d, six times lower than that obtained for the unbridged peptide. However, direct measurement of these dissociation constants by isothermal titration calorimetry shows that the K_d for the C-terminal lactam-bridged peptide is only 2.3 times lower than that for the unsubstituted sequence (B. Wu and J. W. Taylor, unpublished). These data indicate that the difference in the free energies for formation of

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

these two peptide–DNA complexes, $\Delta\Delta G$, lies in the range of 0.5–1.1 kcal/mol, with the lower value probably being more reliable. If this value is interpreted as the helix stabilization achieved by substitution of the lactam-bridge constrained pentapeptide, *cyclo*(1–5)–[Lys–Ala–Ala–Ala–Asp], it is in the same range as the 0.7 kcal/mol value estimated earlier by direct measurement of the helix contents of the dicyclic 14-residue model peptides.[62] Again, different values can be expected, because of differences in the residues that are substituted by introduction of the lactam-bridge constraint. Furthermore, helix stabilization in the GCN4br study also results from the effect of substituting three alanine residues into the GCN4 sequence in place of the native sequence. The helix stabilization that is due to the bridge substitution alone remains to be determined.

In principle, determining a free energy value for the helix stabilization due to a single lactam bridge substituted into an isolated α-helical peptide is most simply achieved by coupling the helix–coil transition to a binding interaction, as in the GCN4 study described above. This converts the helix–coil equilibrium into a two-state reaction, which is more readily fit to a thermodynamic model than is the multistate helix–coil equilibrium of an isolated peptide. A similar approach has been adopted by the group of Hodges,[69] who have studied α-helical coiled-coil formation using urea denaturation to assess the stability of the folded dimeric complex. In this study, two (Glu^i, Lys^{i+4}) residue pairs were substituted into positions 3 and 7 and positions 20 and 24 of a 24-residue peptide sequence that was designed for dimerization to form a coiled coil. The dimerization reaction was then studied for this peptide in its linear form and in its dicyclic (Glu, Lys) lactam-bridged form. Urea denaturation curves, plotted as ΔG for the unfolding reaction *vs* the urea concentration, may be extrapolated back to a urea concentration of zero. This method gave a value for the increased stability of the dimeric complex due to the two lactam bridges, $\Delta\Delta G$, of only 0.3 kcal/mol, or 0.15 kcal/mol per bridge, although strong helix stabilization had been reported earlier for these (Glu^i, Lys^{i+4}) bridges.[61] Alternatively, the midpoints in the urea denaturation curve may be compared in order to eliminate the uncertainties of the extrapolation. This method gave a value of 0.8 kcal/mol per bridge for the helix stabilization, which was more consistent with expectations based on the earlier studies. Differences in the slopes of the extrapolated curves (*m* values) between the bridged and the unbridged structures give rise to concerns that the dicyclic analog may be folded somewhat differently from the unbridged peptide in the dimer. Nevertheless, this dimerization reaction, like the GCN4br-DNA binding reaction, may be a useful cooperative helix–coil transition for comparison and evaluation of the helix stabilizing effects of different lactam bridges.

Table I Mean Residue Ellipticities of κ-Opioid Receptor Peptides Measured at 25°C in 20% HFIP in Water

	Mean Residue Ellipticities (deg · cm^2/dmol)		
Peptide	$[\theta]_{192}$	$[\theta]_{208}$	$[\theta]_{222}$
TMH2	25,000	−10,000	−9,800
A_3-TMH2	59,700	−24,800	−24,300
R_3-TMH2	41,800	−19,100	−17,700
S_3-TMH2	31,400	−15,000	−14,500

The GCN4br study described above raised the question of whether or not there are special requirements for the three intervening residues between the bridging positions in an (i, i + 4)-bridged peptide that determine helix stabilization. For example, the presence of the bridge might require that the side chains of the residues in the first and third intervening positions (residues i + 1 and i + 3) are short, so that their conformations are not restricted by the lactam bridge in the helical conformation, thereby destabilizing the helix. We chose to investigate this question further[62,70] in a study of the hydrophobic second trasmembrane domain (TMH2) of the δ-opioid receptor (residues 81–108).[71] In this study, the CD spectrum of TMH2 in 20% HFIP was compared to those of three analogs in which N-terminal extensions with different (Lys^i, Asp^{i+4}) lactam-bridged sequences were added, as follows:

TMH2: Acetyl–[δ-receptor residues 81–108]–amide
A_3-TMH2: Acetyl–(*cyclo*-1–5)–[Lys–Ala_3–Asp–(δ-receptor residues 81–108)]–amide
R_3-TMH2: Acetyl–(*cyclo*-1–5)–[Lys–Arg_3–Asp–(δ-receptor residues 81–108)]–amide
S_3-TMH2: Acetyl–(*cyclo*-1–5)–[Lys–Ser_3–Asp–(δ-receptor residues 81–108)]–amide

The CD spectra measured for all three lactam-bridged peptides were indicative of an increased helix content compared to that for the unextended TMH2 peptide, as indicated by their larger positive ellipticity peaks at 192 nm and their larger negative ellipticity peaks at 208 and 222 nm (Table I). However, the magnitude of the helix-stabilizing effect was strongly

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

dependent on the sequence of the residues incorporated in between the bridging Lys^1 and Asp^5 residues. An Ala_3 sequence gave the most effective helix stabilization, followed by Arg_3, and then Ser_3. This is the order expected on the basis of the measured helix-stabilizing effects of these residues in linear host–guest systems.[65,72] On the basis of this partial study, therefore, there is no evidence for special requirements for these intervening residues for compatibility with the helix-stabilizing effects of an *i*-to-(i + 4) side-chain lactam bridge. In particular, the relatively large Arg side chain, which is also branched at the guanidino group and charged, appears to be readily accommodated in the bridged helical structure. This does not rule out the possibility of specific interactions with the lactam bridge that might disfavor the helical conformation by stabilizing a particular non-helical conformation. Indeed, we have proposed that the apparent helix-*destabilizing* effect of the (Asp^{17}, Lys^{21}) side-chain lactam bridge placed in hCT (Figure 8) is a result of the *stabilization* of a type I β-turn conformation in residues 17–20.[33] Stabilization of this β-turn may result, at least in part, from hydrogen-bond formation between the bridging Asp^{17} carbonyl and the backbone amide N—H of residue 19. In support of this hypothesis, a type I β-turn formed by residues 17–20 is the only organized conformation identified by proton NMR in the native hCT structure in 85% DMSO in water,[73] where the Asn^{17} side-chain carbonyl may fulfill the same H-bonding role.

There are several reports of peptides incorporating two and even three (i, i + 4)-spaced, Lys-to-Asp or Asp-to-Lys side-chain lactam bridges into nonoverlapping positions in the peptide chain.[37,38,60,69,74] While these peptides typically have a high α-helix content in aqueous solution, the relatively large ring size of these macrocyclic constraints allows them to retain a significant degree of conformational flexibility. Even a 21-residue peptide incorporating three (Lys^i, Asp^{i+4}) side-chain lactam bridges, placed in separated positions in its structure, was found to undergo significant melting over the temperature range from 0 up to 90°C.[38] In contrast, the placement of two (Lys^i, Asp^{i+4}) side-chain lactam bridges in overlapping positions in a peptide consisting of just six amino acid residues has been shown by CD and proton NMR to generate a very stable and regular α-helical structure that is highly resistant to thermal denaturation.[64] The remarkable stability of this α-helical structure is demonstrated in several ways. First, the protected bicyclic hexapeptide, Boc–(*cyclo*-1-5, 2-6)–[Lys–Lys–Ala–Ala–Asp–Asp]–OPac, was shown by detailed proton NMR analysis to adopt an α-helical conformation at low temperature in 50% TFE, despite the fact that such a short peptide structure can form only two backbone amide bonds of the CO^i, NH^{i+4} type that characterize the α-helical conformation. Second, the helical conformation of this protected hexapeptide is shown by CD spectropolarimetry to be altered very little by a change in solvent from 50% aqueous TFE to purely aqueous solution (Figure 10A). Third, proton–deuterium exchange rates for the hydrogen-bonded backbone amides of Asp^5 and Asp^6 in the bicyclic hexapeptide in 50% TFE are extremely slow (Table II). Fourth, CD spectra measured in aqueous buffer at temperatures from 0 up to 80°C indicate that the helical conformation that was observed in detail by proton NMR is almost completely resistant to heat denaturation (Figure 10B).

A decapeptide incorporating the same bicyclic hexapeptide in the central positions, acetyl-(cyclo 3-7, 4-8)-[Arg-Asn-Lys-Lys-Ala-Ala-Asp-Asp-Ser-Arg]-amide, was also recently reported to be fully α-helical in aqueous solution (Figure 11), and demonstrated a similar resistance to heat denaturation.[45] Again, extemely slow proton–deuterium exchange rates were observed for the central backbone amide bonds (Table II), and CD spectra measured at temperatures from 0 up to 80°C all indicated a similar high helix content. This study extended the prior characterization of the protected bicyclic hexapeptide in two important ways. First, it demonstrated that a regular α-helical structure is adopted by the bicyclic (Lys^i, Asp^{i+4})-bridged structure in aqueous solution without TFE addition; second, it demonstrated that this central helical structure can act as an α-helix initiator, propagating the α-helical conformation through unconstrained residues that have been added to its N-terminal end or to its C-terminal end. It is interesting to note that the decapeptide, like the hexapeptide, also melted very little over the entire temperature range studied,[45] despite the fact that four of the ten residues in its structure lie outside of the bicyclic core, and are unconstrained. This is consistent with the core, helix-initiating, bicyclic structure being essentially fixed in the α-helical conformation across the entire temperature range studied, and with helix propagation by each unconstrained residue being a relatively probable event that is entropically controlled and largely independent of temperature. These considerations highlight the potential utility of the core (*cyclo*-1–5, 2–6)–[Lys–Lys–Ala–Ala–Asp–Asp] structure as a peptidomimetic for α-helical structures in proteins. This structure should prove to be useful for studying protein folding from an α-helical core, or protein recognition involving α-helical surfaces, or for performing experiments on rigid α-helical scaffolds. All of these potential applications are now rendered much more

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

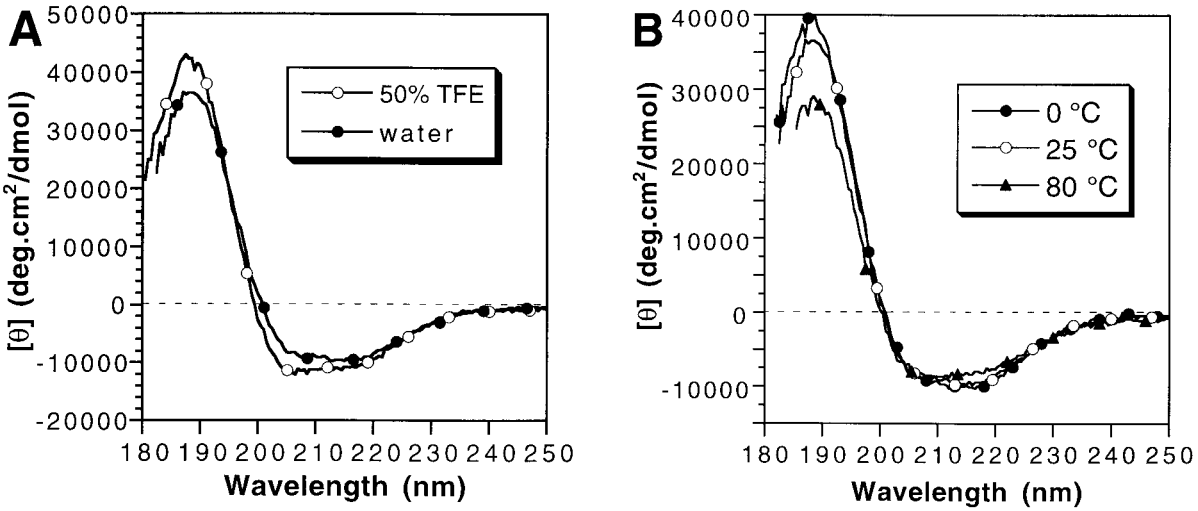


FIGURE 10 Stability of the α-helix formed by a bicyclic (Lys^{i}, Asp^{i+4})-bridged hexapeptide. CD spectra measured for solutions of the protected peptide Boc–(*cyclo*-1–5, 2–6)–[Lys–Lys–Ala–Ala–Asp–Asp]–OPac in aqueous buffer, pH 7.0, and in 50% TFE, at 25°C are compared in panel A. CD spectra measured for the aqueous peptide solution at different temperatures are compared in panel B.[64] These spectra demonstrate that the α-helical conformation of this peptide is insensitive to changes in solvent or temperature.

accessible through the synthetic approach to this bicyclic peptide as a building block for segment condensation peptide assembly that was described above (Figure 4).[51]

Stabilization of the α-Helical Conformation Using (*i*, *i* + 7) Side-Chain Lactam Bridges

The side chains of a pair of amino acid residues that are separated by six intervening residues in the primary structure of a peptide are aligned on the same side of an α-helix, and are almost exactly two helical turns apart (see Figure 6). Several groups have, therefore, considered it worthwhile to explore the helix stabilizing effect of linking such residue pairs *via* their side chains, creating (*i*, *i* + 7) side-chain bridges. These bridges must pass over or between the side chains of the residues in positions *i* + 3 and *i* + 4. In the first reported example, from the Schultz group,[75] effective helix stabilization was attained using a disulfide link to connect a D-thiolysine residue (Lys with a thiol-functionalized side chain instead of the ε-amine) in the N-terminal, *i*th position, to the corresponding L-thiolysine residue placed in the C-terminal, (*i* + 7)th position. This bridge design takes advantage of the fact that the C$^{\alpha}$—C$^{\beta}$ side-chain bonds in the bridging N-terminal D-amino acid and C-terminal L-amino acid residues will be oriented toward each other, simplifying the folding requirements for the bridge. In contrast, the L-Lys side chain in a (Lys^{i}, Asp^{i+4}) lactam bridge, for example, has its C$^{\alpha}$—C$^{\beta}$

Table II Amide Hydrogen–Deuterium Exchange Rates in Bicyclic Peptides

	Exchange rate constant, $k_{ex} \times 10^5$ (min^{-1})[a]									
Peptide amides	Arg^1	Asn^2	Lys^3	Lys^4	Ala^5	Ala^6	Asp^7	Asp^8	Ser^9	Arg^{10}
Hexapeptide backbone			6800	>1000	210	24	<0.1	160		
Hexapeptide bridges			>1000	>1000						
Decapeptide backbone	3800	4600	83	13	3.9	1.8	<1	<1	430	1800
Decapeptide bridges			2200	2200						

[a] Residues are numbered according to the bicyclic decapeptide sequence.
[b] Measured in 50% TFE–water, pH 3.7, 22°C.[64]
[c] Measured in water, at pD 2.6, 5°C.[45]

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

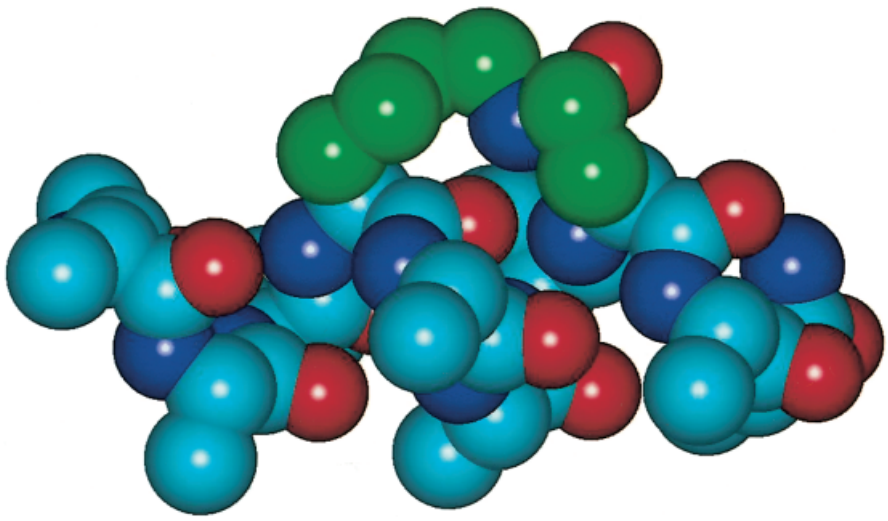


FIGURE 11 Structure of a bicyclic (i, $i + 4$)-bridged α-helical peptide determined by ^{1}H-NMR and distance-constrained molecular dynamics. The structure shown is an average of six highly converged low-energy conformers of the decapeptide Ac–(*cyclo*-3–7, 4–8)–[Arg–Asn–Lys–Lys–Ala–Ala–Asp–Asp–Ser–Arg]–NH_2 that were obtained by distance-constrained annealing.[45] The side chains of residues 1, 2, 9, and 10, which are unconstrained and not well converged, are represented by methyl groups, for clarity. The CPK model is arranged with its N-terminal end to the left, and shown with the carbon atoms of the (Lys^4, Asp^8) side-chain lactam bridge highlighted in green. The second bridge is hidden from view. The H-bond formed by the bridging amide and the Lys^4 backbone carbonyl, as well as several backbone H-bonds, are clearly visible.

bond oriented toward the N-terminus, so that the γ, δ, and ε methylene groups are constrained to fold back towards the C-terminus of the helix. Helix-disrupting effects of the N-terminal D-thiolysine residue in this i, $i + 7$ disulfide bridge were reported for a monocyclic apamin-derived peptide in which the bridge was used and studied in detail by NMR.[76] This illustrates the potential for helix destabilization that can result from incorporating a D-amino acid residue in a helical structure.[77] Nevertheless, the combination of D-and L-residues in side-chain bridge design for helix stabilization might usefully be explored further, and may find applications where N-terminal helix termination is tolerated or even desired.

Phelan et al.[76] have investigated the side-chain linkage of (i, $i + 7$)-spaced glutamic acid residues by amide-bond formation with 1,5-diaminopentane. This bridge consists of nine methylene groups and two amide bonds, for a total of 13 atoms linking the two backbone α carbons in the ith and ($i + 7$)th positions. The flexibility of this bridge on a helix surface is, therefore, comparable to that of the 10-atom disulfide bridge described above. This lactam bridge was placed into positions 3 and 10 of an N^{α}-acetylated, 13-residue, alanine-rich model peptide amide and generated a highly α-helical structure that was characterized in detail by ^{1}H-NMR and CD, and appeared to be more effective for helix stabilization than the disulfide bridge of Jackson and co-workers.[75]

We have also tested lactam bridges linking (i, $i + 7$)-spaced residues,[62] using alanine or β-alanine residues to bridge the side chains of Lys in the third position and Asp in the 10th position of our 14-residue model peptide (Figure 6). Three monocyclic peptides were tested, one incorporating a $Lys(Ala)^3$-to-Asp^{10} 10-atom bridge, a second incorporating a $Lys(\beta\text{-}Ala)^3$-to-Asp^{10} 11-atom bridge, and a third incorporating a $Lys(\beta\text{-}Ala)^3$-to-Glu^{10} 12-atom bridge. Surprisingly, the CD spectra of these peptides indicated that none of them had any helical structure in aqueous solution. In 50% TFE, the model peptides with the 11-atom and 12-atom bridges were significantly helical, but the CD spectrum for the peptide having the 10-atom $Lys(Ala)^3$-to-Asp^{10} bridge was still indicative of mostly disordered conformations.

A consideration of all of the studies of (i, $i + 7$)-bridged peptides described above prompted us to begin exploring alternative i, $i + 7$ helix-stabilizing bridge designs. The overall goal was to identify relatively rigid linking structures, rather than flexible chains consisting primarily of methylene groups, to connect two (i, $i + 7$)-spaced L-amino acid residues using amide bonds. A series of four monocyclic peptide analogues that incorporated isomeric bridging

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

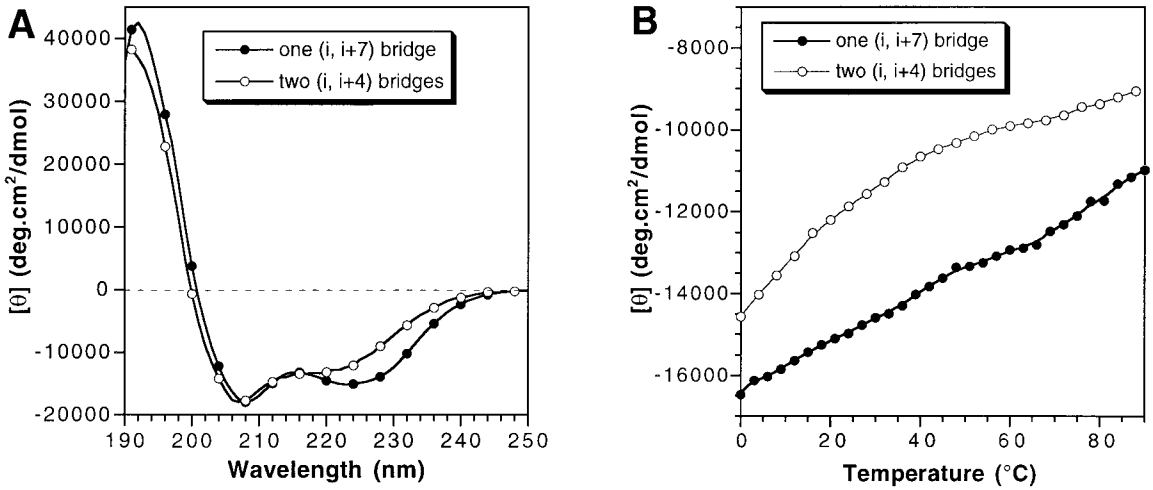


FIGURE 12 Comparison of the helix-stabilizing effects of incorporating one helix-stabilizing (i, i+7) lactam bridge or two helix stabilizing (i, i+4) lactam bridges into the same model peptide sequence. CD spectra (A) and CD melting curves (B) were measured for the dicyclic (Lys^{i}, Asp^{i+4})-bridged 14-residue peptide shown in Figure 6A and the corresponding monocyclic (Dap[AMPA]^{i}, Asp^{i+4})-bridged peptide shown in Figure 6B. Both peptides were studied in phosphate buffer at pH 7.0. Spectra were measured at 25°C, and the melting curves were measured by following [θ] at 222 nm.

structures based on *para*-substituted benzene rings into residue positions three and ten of the earlier 14-residue model peptide sequence was reported.[32] The most effective helix stabilizing bridge consisted of a 1,2-diaminopropionic acid residue (Dap) in position 3 linked via a *p*-(aminomethyl)phenylacetic acid residue (AMPA) to an aspartic acid residue in position 10. The structure of this peptide is illustrated in Figure 6B. Based on the CD spectra in aqueous solution, just one (Dap[AMPA]^{3}, Asp^{10}) bridge was found to be at least as effective for helix stabilization as two Lys^{i}, Asp^{i+4} bridges placed in the same 14-residue model peptide, and to endow the helical structure with a somewhat greater resistance to heat denaturation (Figure 12). Subsequently, a detailed proton NMR study[78] showed that a regular α-helical structure was formed by this monocyclic AMPA-bridged peptide throughout the bridged sequence, except for a small distortion around residue 9 that seems to accommodate an H-bond formed between the side-chain amide link to Asp10 in the bridge and the backbone CO of Lys^{6}. A detailed examination of 11 low-energy structures generated by nuclear Overhauser effect (NOE)-distance-constrained, simulated annealing indicated that the backbone conformations from residue 2 through residue 12 were all essentially in the α-helical region of ϕ–ψ space, with ϕ values in the range between $-40°$ and $-100°$, and ψ values in the range from $-10°$ to $-60°$ (Figure 13).

The (Dap[AMPA]^{i}, Asp^{i+7}) bridge has also been tested in bicyclic structures. Two (Dap[AMPA]^{i}, Asp^{i+7}) bridges placed in our 14-residue model peptide structure (Figure 6); such that they overlap to link residues 3 and 10 and residues 7 and 14, give rise to a highly α-helical structure. This structure shows a resistance to heat denaturation that is similar to that already described for the bicyclic (Lys^{i}, Asp^{i+4})-bridged peptides (Figure 14).[48,62] Other combinations of bridges and other bridge placements in bicyclic peptides may also give rise to similar denaturation-resistant helical structures. For example, combining a (Dap[AMPA]^{i}, Asp^{i+7}) bridge linking residues 3 and 10 in our 14-residue model peptide with a (Lys^{i}, Asp^{i+4}) bridge linking residues 4 and 8 also gives a stable helical structure, according to its CD spectra. However, the phenomenon is not completely general, since we have identified two nine-residue bicyclic peptides with similar combinations of overlapping bridges that are clearly nonhelical, judging from their CD spectra.[62]

Conformational Constraints Using (*i*, *i* + 3) Side-Chain Lactam Bridges

Two (Asp^{i}, Lys^{i+3}) side-chain lactam bridges have been incorporated into an amphiphilic α- or π-helical human apolipoprotein E (apoE) peptide, and these constraints have been determined to stabilize α-heli-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

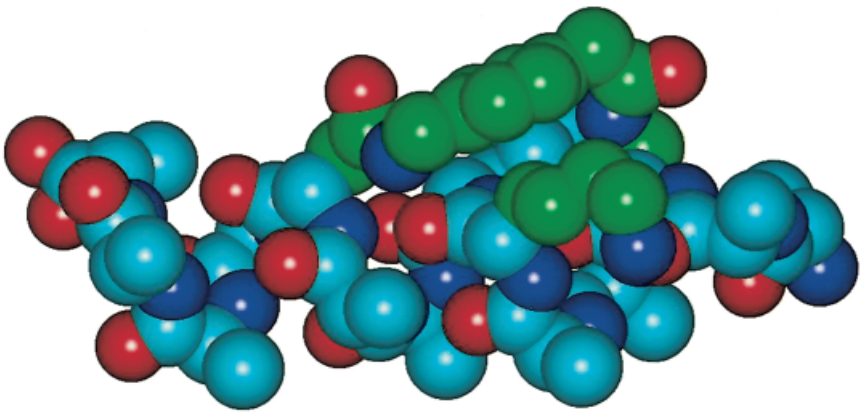


FIGURE 13 Structure of a monocyclic (*i*, *i* + 7)-bridged α-helical peptide determined by ^{1}H-NMR and distance-constrained molecular dynamics. The structure shown is a representative low-energy conformer of the (Dap[AMPA]^{i},-Asp^{i+7})-bridged monocyclic 14-residue peptide shown in Figure 6B that was obtained by distance-constrained annealing.[78] With the exception of Lys^{6} and Gln^{7}, the side chains of all of the unbridged residues are represented by methyl groups, for clarity. The CPK model is arranged with its C-terminal end to the left, and shown with the carbon atoms of the (Dap[AMPA]^{3}, Asp^{10}) side-chain bridge and of the Lys6 side chain highlighted in green. The side-chain amide H-bond to the Lys^{6} backbone carbonyl is clearly visible, and the orientation of the Lys^{6} methylene side chains across the face of the bridging phenyl ring is apparent. The edge-on orientation of the Gln^{7} side chain with respect to the bridging phenyl ring is hidden from view.

cal structure relative to the native peptide, using CD and proton NMR.[79] In this case, the 24-residue peptide, H–Gly–[apoE(41–62)]–Gly–OH, was constrained by placement of an (Asp^{i}, Lys^{i+3}) bridge in residue positions 2–5 and 19–22, in place of a (Gln, Ser) pair at the N-terminus and a (Gln, Arg) pair at the C-terminus. The helix melting curve of this dicyclic peptide compared to that of the corresponding native peptide was followed using the CD signal at 208 nm, and then a van't Hoff analysis of this data was performed, assuming a cooperative, two-state helix–coil transition. This study was undertaken using TFE (50% v/v) as cosolvent, in order that both peptides would have sufficient helical structure to be measured by the CD method. From this analysis, the two bridges in the dicyclic analog were determined to result in a total of about 1 kcal/mol of helix stabilization relative to the linear analog. Parallel lines in the van't Hoff plot for the melting transitions of the two peptides indicated that the stabilizing effect of the bridges was entirely entropic, as anticipated for a conformational constraint. Based on these measurements, the (*i*, *i* + 3)-bridge is, therefore, about as effective for helix stabilization as the Lys^{i}, Asp^{i+4} bridge discussed earlier. However, the use of TFE as a cosolvent for the CD melts calls into question the cooperativity of the helix–coil transition, as TFE is known to dramatically reduce the cooperativity of this transition in linear Ala-rich peptides.[80] Indeed, the published data for the van't Hoff analysis of the dicyclic apoE peptide shows a linear increase in % helix with increasing temperature within the range of accessible conditions (0–70°C), suggesting that a multistate model for the helix unfolding is a more appropriate fit. Interestingly,

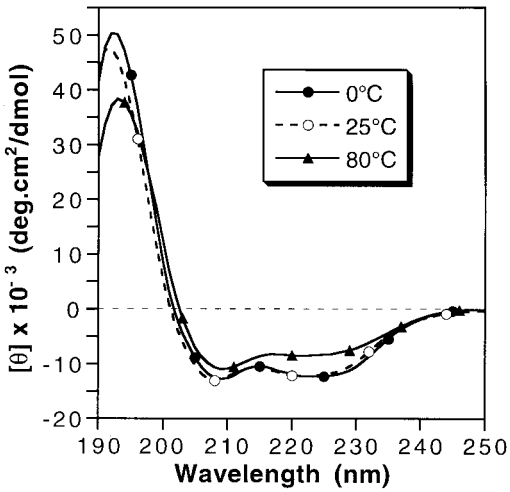


FIGURE 14 Stability of the α-helical structure of a bicyclic (*i*, *i* + 7)-bridged peptide with respect to thermal denaturation. CD spectra of the bicyclic peptide H–(*cyclo*-3–10, 7–14)–(Lys–Leu–Dap[AMPA]–Glu–Leu–Lys–Dap-[AMPA]–Lys–Leu–Asp–Glu–Leu–Lys–Asp)–OH were measured in aqueous buffer, pH 7.0, at the temperatures indicated.

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

though, the peptide design in this study, in which lactam bridges are placed at each terminus, preceeds a similar design that we have deliberately pursued in order to generate a peptide that *does* undergo a cooperative helix–coil transition in purely aqueous solvent.[31] The difference between the melting behaviors of these two peptides may be due to differences in their primary structures, or the use of different lactam bridges to stabilize the helix ends.

NMR structure determination of the dicyclic analog of H–Gly^0–[apoE(41–62)]–Gly–OH in 50% TFE was also undertaken by Luo et al.,[79] and showed that two segments of regular α-helical structure were adopted by the dicyclic peptide around the bridged regions, but also indicated a nonhelical bend or kink in the middle of the structure. This suggested to the authors that there might be some hydrophobic packing between the helical segments or at the link between them. These effects may also give rise to helix stabilization, although no corresponding long-range NOE signals were observed. Despite the concerns about the lack of cooperativity and the potential for intramolecular hydrophobic packing of the helical segments, this study does suggest that (Asp^i, Lys^{i+3}) side-chain lactam bridges are worthy of further study for the purpose of stabilizing α-helical structure.

In contrast to the (Asp^i, Lys^{i+3}) side-chain lactam bridges described above, Houston et al.[61] have reported that (i, i + 3)-spaced lysine and glutamic acid residues, linked in either orientation, actually destabilize helical structure, as judged by CD measurements. Scievano et al.[81] have recently taken advantage of this observation, in their design of peptide models that adopt the 3_{10}-helical conformation in the presence of SDS micelles. In this study, α-aminoisobutyric acid residues (Aib) were introduced into a model peptide system in order to bias the conformation in favor of either the 3_{10}-helical *or* the α-helical conformation, and then (Glu^i, Lys^{i+3}) side-chain lactam bridges were used in combination with Aib to disfavor the α-helical conformation. A series of peptides with the structure acetyl–([cyclo-1-4]–[Glu–Aib–Aib–Lys])$_n$ amide, in which n = 1, 2 or 3, were synthesized and studied by CD and NMR. CD studies in 1 m*M* SDS showed that, as the peptide chain length increased from one repeat sequence to two and three, there was an increase in the fractional content of a helical conformation with the spectral characteristics of a regular α-helical structure (a maximum below 195 nm, and a minumum at 208 nm with a pronounced shoulder around 222 nm). In the proton NMR spectra, however, pronounced NOE connectivities between the αH^i *and* NH^{i+2} and the αH^i and NH^{i+3} proton pairs, indicated increasing 3_{10}-helix content as the peptide chain length was increased. This conclusion was supported by NOE-distance-constrained molecular modeling of the trimer sequence, which showed the formation of a 3_{10}-helix through residues 1–8. The stabilizing effect of the (Glu^i, Lys^{i+3}) side-chain bridges on the 3_{10}-helical structure remains to be established, however, since the conformational properties of linear, isosteric analogs have not been determined for comparison and the importance of conformational stabilization by the SDS is unknown. Furthermore, the utility of these bridges for *disfavoring* formation of an α-helix (Houston et al., 1995) has been called into question by CD studies of (Glu^i, Lys^{i+3})-bridged CRF analogs, which indicate that this bridge type is fully compatible with the α-helical conformation, even if it is not helix stabilizing.[82] Nevertheless, this study demonstrates an effective approach to designing a constrained, 3_{10}-helical peptide.

Naider and co-workers have been interested in applying (i, i + 3) side-chain lactam bridges to the constraint of turn structures, in order to investigate the bioactive conformation of the yeast pheromone 13-residue peptide, α-factor.[83] Yeast α-factor has a Lys^7–Pro^8–Gly^9–Gln^{10} sequence, which was anticipated to favor a type II β-turn conformation directed by the Pro–Gly sequence that might represent its receptor-bound conformation. The Naider group has undertaken a detailed study of the effects of side-chain composition and ring size on the conformation of model tetrapeptides having the general structure acetyl–(*cyclo*-1-4)–[Xxx–Pro–Gly–Yyy]–amide, in which side-chain lactam bridges linked all combinations of residues Xxx = Dap, Dab, Orn, or Lys to residues Yyy = Glu or Asp. For conformational analysis, molecular dynamics simulations were performed in a vacuum and in simulated DMSO solution, and the calculated conformations were compared to rotating frame nuclear Overhauser effect spectroscopy (ROESY)-derived distances from the NMR spectra measured in DMSO-d_6. A variety of turn conformations were identified based on agreement between the modeling predictions and the NMR experiments. These included type II β-turns, which were observed for the (Dap, Asp)-, the (Dab, Glu)-, and the (Lys, Asp)-bridged peptides, and β- or γ-turns observed for the (Orn, Asp)- and (Orn, Glu)-bridged peptides. The largest ring constraint, from a (Lys, Glu) bridge, resulted in a flexible structure with no strong structural preference. Some of these bridged tetrapeptides have been incorporated into yeast α-factor and examined for their solution conformations in DMSO-d_6, and their effects on the biological activities of the pheromone in aqueous media.[84] Interestingly, the confor-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

mational preferences observed for the tetrapeptides alone were apparently retained in the context of the full-length pheromone. This suggests that such (*i*, *i* + 3)-bridged Pro^{i+1}-Gly^{i+2} tetrapeptide structures, may have further utility as building blocks controlling the turn conformations of other "host" peptide sequences.

FUNCTIONAL ACTIVITIES OF PEPTIDES INVESTIGATED USING SIDE-CHAIN LACTAM BRIDGE CONSTRAINTS

Models for Protein Folding

There has been considerable interest in applying conformationally constrained peptides or peptidomimetics to studies of protein folding. Helix nucleating cyclic peptidomimetics that place H-bonding carbonyls in the appropriate geometry,[85,86] and monocyclized peptides incorporating backbone H-bond surrogates,[59,87] have been used to study propagation of α-helical structure in the C-terminal direction. Rigid β-turn mimetics have been used to nucleate and study antiparallel β-sheet formation from the strand ends,[88] and H-bonding templates for the lengthwise stabilization of the first strand of a β-sheet have also been proposed.[89–91]

The α-helical structure of the bicyclic lactam-bridged hexapeptide sequence (*cyclo*-1-5, 2-6)–[Lys–Lys–Ala–Ala–Asp–Asp]–, and its apparent persistence over a wide temperature range (Figures 10 and 11), have led us to use this structure as a model or surrogate for α-helices of biological significance. This approach allows, for example, studies of helix propagation in protein structures that are not readily accessible for direct observation. In the first such study (L. Xie and J. W. Taylor, unpublished), this bicyclic helix initiator was used to explore the conformational propensities of the fourth transmembrane domain (TMD4) predicted for the κ-opioid receptor, and its extension into the second extracellular loop sequence (EL2). This region of the receptor contained within its sequence the dipeptide–Gly–Gly– in residue positions 197 and 198. This sequence had variously been predicted to terminate a helical structure in TMD4 at its C-terminal end at the extracellular membrane/water interface,[91] or to support propagation of the TMD4 helix within the hydrophobic core of the membrane.[92,93] In order to test whether or not two consecutive glycine residues could support helix propagation into the κ-EL2 peptide sequence, two helix-initiated peptides were synthesized. In one peptide, the bicyclic lactam-bridged helix initiator was connected directly to the κ-EL2 sequence, corresponding to κ-receptor residues 199–209 (κ-EL2G0). In another peptide, the helix-initiating peptide was connected to κ-EL2 *via* two glycine residues corresponding to the Gly^{197}–Gly^{198} sequence (κ-EL2G2). The κ-EL2 peptide alone was also synthesized for the purpose of comparisons. CD spectra measured for these peptides in 45 or 80% HFIP at 25°C indicated that both of the helix-initiated peptides could attain a similar, high fractional α-helix content, whereas the κ-EL2 peptide alone had much less α-helical structure. This study demonstrated that propagation of a stable helix in the C-terminal direction through consecutive glycine residues and into the κ-EL2 sequence was possible, at least in a hydrophobic environment, and that helix propagation through Gly^{197}–Gly^{198} should be considered in modeling the κ-opioid receptor.

In a similar application of the (Lys^{i}, Asp^{i+4}) lactam-bridged bicyclic hexapeptide, helix propagation has been studied in a region of gp41, the major transmembrane glycoprotein on the surface of human immunodeficiency virus type 1 (HIV-1), that has proven inaccessible to direct structural determination.[51,54] Helical segments of gp41 are thought to undergo significant conformational rearrangements on the pathway leading to viral infection of $CD4^+$ cells.[94] In this study, the helix initiating bicyclic peptide was used to mimic the C-peptide helix from the gp41 ectodomain,[95,96] in order to study the potential for propagation of this helix in its C-terminal direction through the adjoining, membrane-proximal gp41 sequence. The bicyclic hexapeptide was linked to eight residues corresponding to the gp41 sequence immediately C-terminal to the C peptide helix. These residues incorporate the recognition epitope, centered around the sequence Glu–Leu–Asp–Lys–Trp–Ala–Ser, for the HIV-1 neutralizing antibody, 2F5.[97] This epitope adopts a type I β-turn conformation in the Asp–Lys–Trp–Ala sequence, in the crystal structure described for the 2F5-bound complex.[98] However, helix initiation by N-terminal addition of (*cyclo*-1-5, 2-6)–[Lys–Lys–Ala–Ala–Asp–Asp]– stabilized a high fractional helix content in this epitope. This suggests that the native gp41 conformation in this undefined region may be α-helical, at least when the C-peptide helix is formed. Binding of the epitope to the monoclonal antibody, 2F5, in a β-turn conformation would then require an induced fit of the gp41 peptide, and this conformational change may be of functional importance in determining the HIV-1 neutralizing properties of the antibody.

The helix stabilizing lactam loops formed by linking Lys^{i} and Asp^{i+4} side chains have also been ap-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

plied to the development of a model peptide system that allows detailed calorimetric studies of the thermodynamics of the folding and unfolding of an α helix.[31] The helix–coil transition cannot readily be studied using an isolated α-helical structure in an unconstrained monomeric peptide: the α-helix is normally only marginally stable relative to disordered conformations, and the folding transition is noncooperative and only partially complete in the accessible temperature range of aqueous solutions (essentially 0 to 100°C). Calorimetric analyses of the helix–coil transition have, therefore, generally been limited to indirect studies based on the effects of mutations on the denaturation of globular proteins, or of simpler cooperatively unfolding structures such as α-helical coiled coils.[99] Attempts to study monomeric peptides that form isolated α-helical structures have generally been limited to a partial analysis of the folding of alanine-rich peptides by calorimetry,[100] residue-by-residue analyses of helix melting by ^{13}C-NMR,[101] or analyses of CD melts performed in TFE solutions and extrapolated back to water.[80]

In the process of studying the effects of helix stabilization on DNA binding by the basic region peptide from GCN4 (GCN4br),[67] we discovered that a dicyclic 29-residue analog with Lys^{i} and Asp^{i+4} side-chain lactam bridges placed at each of its termini underwent a cooperative helix–coil transition that was nearly complete between 0 and 90°C in water. In contrast, the unbridged linear analog showed only a low helix content, even at 0°C, and showed a broad thermal transition to the unfolded state. This comparison is illustrated in Figure 15, using the melting curves obtained by CD spectropolarimetry. The cooperative nature of the helix–coil transition in the bridged peptide was explored in detail by differential scanning microcalorimetry.[31] This study showed that the excess and partial molar heat capacity functions of the dicyclic GCN4br analog fit a two-state transition, and that the fractional helix content of this peptide ranged from >95% at −10°C to about 1% at 90°C. This analysis, therefore, gave a compete calorimetric description of the helix–coil transition for this peptide, without the need for extended extrapolations of the data. This analysis was precise enough to allow the determination of a reproducible temperature-independent value for the heat capacity, ΔC_p, for the unfolding transition, in addition to temperature-dependent calorimetric enthalpy and entropy functions, ΔH and ΔS. At 30°C, the enthalpy value for the unfolding reaction was 650 ± 50 cal mol^{-1} per residue, which exactly matched the value determined previously for a helical coiled-coil peptide.[99] Since the folding of the coiled-coil peptide depends on the burial of nonpolar surfaces, enthalpic contributions due to nonpolar interactions were assumed to be zero at this temperature, and this helix-stabilizing enthalpy could be assigned to the formation of backbone hydrogen bonds, exclusively. This value also agreed well with the earlier ^{13}C-NMR study of small Ala-rich peptides by Shalongo et al.[101]

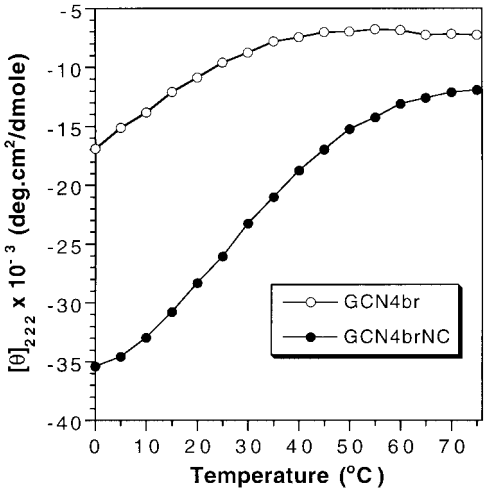


FIGURE 15 Cooperature melting of a dicyclic GCN4 basic-region peptide with helix stabilizing lactam bridges at both termini. CD melting curves for the α-helical structures formed in phosphate-buffered saline, pH 7.4, by a 29-residue, GCN4 basic region peptide (GCN4br), and an analogue of this peptide with (Lys^{i}, Asp^{i+4}) side-chain lactam bridges incorporated into each terminus (GCN4brNC), are compared. The linear, unbridged peptide was only partially helical at 0 °C and showed a broad melting transition, whereas the dicyclic peptide was about 95% helical at 0 °C and melted in a cooperative fashion that fit thermodynamic models for a two-state thermal denaturation.[31]

A small positive value was determined for ΔC_p for unfolding of the isolated α-helical GCN4br peptide. Again, this result agrees with the prior residue-specific analyses of Ala-rich peptides by ^{13}C-NMR. Since the peptide backbone consists predominantly of polar groups, which are associated with negative values for ΔC_p for unfolding, these results are indicative of significant positive contributions to the ΔC_p value arising from the exposure to water of nonpolar surfaces with unfolding.[102] This shows that stabilization of an isolated α-helix due to the burial of nonpolar surfaces can be significant. However, further analysis will be needed to determine whether these contributions arise from backbone interactions predominantly, or from the packing of nonpolar side chains on the helix surface. The packing of lysine side chains onto the surface of a polyalanine helix, which would in-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

clude a significant nonpolar component, has previously been determined to have a strong helix stabilizing effect.[103] The dicyclic GCN4 peptide used in this calorimetric study is also Ala rich, and has numerous Arg (seven), Lys (two), and Leu (one) side chains that might produce similar helix-stabilizing effects, thereby contributing to the positive ΔC_p value observed. The cooperative folding exhibited by this helical dicyclic peptide should provide an excellent model system for addressing this and other fundamental questions relating to the folding and stabilization of isolated α-helices.

Lactam-Bridged Peptides Designed for Increased Receptor Binding Affinity

In cases where a peptide ligand is largly disordered in aqueous solution, but is known to adopt an α-helical conformation in order to express its biological activity through a binding interaction, helix stabilization by conformational constraint should lead directly to increased potency. Two examples illustrate how lactam bridges that are known to stabilize the α-helical conformation have been used for this purpose. In the first, already discussed above, the affinity of the basic region of the yeast transcription factor, GCN4 (GCN4br), for its cognate sequence in duplex DNA was studied.[67,68] GCN4br is known to undergo a coil–helix transition upon specific binding into the major groove of the duplex DNA recognition half-site sequence 5′-TGAC-3′. The side-chain lactam bridged pentapeptide (*cyclo*-1-5)–[Lys–Ala–Ala–Ala–Asp] was substituted for a 29-residue native GCN4br sequence at either the N-terminal end (GCN4brN) or the C-terminal end (GCN4brC), or at both ends (GCN4brNC). Since these peptides were designed, using the crystal structure of the peptide–DNA complex, such that the lactam bridged substitutions would not interfere with DNA recognition, helix stabilization by the bridged peptides was expected to translate directly into increased DNA binding affinity. This indeed was the result: the helix contents of the peptides in phosphate-buffered saline at pH 7 were ordered GCN4brNC $>$ GCN4brC $>$ GCN4brN $\approx$ GCN4br; the peptide affinities for the duplex DNA sequence 5′-GCAATGACGAGC-3′ were assessed by isothermal titration calorimetry and followed the same order, giving dissociation constants for the peptide complexes of 0.79 $\pm$ 0.30 μM for GCN4brNC, 1.79 $\pm$ 0.07 μM for GCN4brC, 2.89 $\pm$ 0.28 μM for GCN4brN, and 4.12 $\pm$ 0.07 μM for GCN4br (B. Wu and J. W. Taylor, unpublished).

The second example illustrates how lactam bridges may be used to probe a peptide recognition event, in order to confirm a structure-based model for its conformational requirements. Peptide sequences from the HIV-1 glycoprotein gp41 have long been known as potent inhibitors of the fusogenic activity of the virus.[104,105] The most effective inhibitors have sequence overlap with C peptide from gp41. C peptide, together with N peptide, forms a six-helix bundle upon trimer formation by gp41 constructs.[95,96] This helix bundle is thought to be the stable end product of conformational changes that gp41 undergoes during fusion of the HIV-1 membrane with the membrane of a $CD4^+$ cell. [94] The gp41 peptides that inhibit fusion are largely unstructured in solution, but have been assumed to displace C peptide during the conformational rearrangements of fusion-primed gp41, and to adopt a helical conformation in expressing this activity. McDowell and co-workers[106] have shown that (i, $i + 7$) lactam-bridged analogs of the inhibitory peptides that are properly constrained to be more helical are also more effective inhibitors of cell infection by HIV-1. Moreover, the crystal structure of the helix bundle is consistent with their data, in that conformational constraints that are positioned in the inhibitory peptides such that they would interfere with formation of the six-helix bundle are also disruptive of inhibitory activity. This study provides a direct link, therefore, between the crystal structure of the six-helix bundle and the fusogenic activity of C-peptide-derived gp41 peptides. A structural basis for the design of novel agents for blocking HIV-1 infection is thereby established.

Lactam-Bridged Amphiphilic α-Helical Peptide Hormones

Determining the functional conformations of medium-length bioactive peptides (15–50 residues) that contain, at most, one disulfide bridge, is among the most difficult challenges in peptide science. Such peptides are too small to fold independently in solution into stable globular structures that might represent their bioactive conformations, and the complexes they form with receptors are often inaccessible to direct structural analysis. However, these medium-sized peptides are large enough such that significant elements of secondary and even tertiary structure may be present in their receptor-bound conformations. The incorporation of lactam bridges into the structures of these flexible peptides is an attractive approach to solving this problem. Side-chain lactam bridges, spaced several residues apart in the linear sequence, will constrain segments of the peptide chain such that they have a much smaller set of accessible conformations, and may even favor one conformation in par-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ticular. If a correlation can be established between the stabilization of a particular conformation and an increase in the receptor binding affinity of the peptide ligand, or its agonist or antagonist potency, then the functional importance of that conformation will be indicated. Further rational design based on such an indication may then be performed to test or confirm the functional significance of the conformation that has been identified.

Given the nearly limitless possibilities for the bioactive conformation of a flexible linear peptide in the 15–50 residue size range, the design and placement of a conformational constraint must necessarily be guided by a conformational hypothesis. In their studies of the yeast α-factor mating pheromone, for example, Naider and colleagues were guided in their placement of a side-chain lactam bridge constraint by the hypothesis that the Pro^8–Gly^9 sequence in this peptide would direct a β-turn in the bioactive state of this 13-residue peptide. However, after a preliminary search for turn stabilizing (i, $i + 3$) bridges,[83] as described above, the lactam-constrained analogs were found to have lost considerable receptor binding potency, indicating that the conformational constraints were inappropriate for constraint of the bioactive conformation, or the bridges interfered with receptor recognition.[84]

The hypothesis that amphiphilic α-helical structures may determine the active conformations of many midsized peptide hormones[66,107,108] also provides a hypothesis for guiding lactam bridge design and placement in peptide hormones. Helix stabilizing effects of several bridge designs have been characterized, as described above, and these bridges may be placed within the sequences identified as having the potential to form an amphiphilic α-helix. Assuming that these structures may serve as receptor binding motifs, their amphiphilic character is strongly suggestive of bridge placement on the hydrophilic face of the helices, where receptor recognition is least likely to be disrupted. The earliest example of this approach was that of Felix and co-workers,[28] who adopted a conservative approach to searching for appropriate bridge placements in GRF. These coworkers identified two potential amphiphilic α-helical segments in a human GRF(1–29) analog, [Ala^{15}]–GRF(1–29)–amide, and chose to substitute the residue pair (Asn^8, Lys^{12}) with an (Asp, Lys) side-chain bridge, in the N-terminal helical segment. In the C-terminal helical segment, a similar search for conservative substitutions led to the placement of a (Lys, Asp) bridge in place of Lys^{21} and Asp^{25}. These analogs, and numerous others with altered ring sizes, were studied for their growth-hormone releasing potencies. Overall, the cyclic constraints were determined to be compatible with bioactivity, only if the ring sizes were large enough to be compatible with the α-helical conformation (20-atom rings or larger). However, several analogs with bridges that were ring-expanded (up to 24-atom rings) compared to the (Asp, Lys) and (Lys, Asp) bridges (20-atom rings) had higher potencies. A 22-atom ring size was generally found to be optimal, even though the helix stabilization and constraint in the larger ring-size analogs was less.[29]

Based on these extensive studies of GRF, the requirement for a helical conformation in the N- and C-terminal segments of GRF(1–29) is strongly indicated. We may speculate that the requirement for a larger, more flexible bridge for optimal activity indicates that these helices may be curved or distorted on the receptor surface, or that the shorter bridges are interfering directly with receptor binding or activation. In the latter case, a thorough exploration of the placement of the rings might identify a better choice. Taking these findings into consideration, Gulyas et al.[82] adopted a thorough "scanning" approach toward lactam bridge placement in CRF antagonists, and used more flexible (Glu^i,-Lys^{i+3}) bridges for the constraint. Like GRF, CRF was also suspected of adopting an amphiphilic α-helical conformation in its biactive form. A scan of these (i, $i + 3$) bridges across the whole sequence of a rat/human CRF(12–41) analog identified a bridged antagonist that was 30 times more potent than the unsubstituted analog in an *in vivo* assay for ACTH secretion, and showed a high helix content identical to that of the linear, unsubstituted analog in TFE. A similar scanning approach has been suggested by the Hruby group,[44] who have scanned the glucagon sequence using (Lys^i, Glu^{i+4}) side-chain lactam bridges. Glucagon has two potential amphiphilic α-helical segments located within its 29-residue structure. Two correspondingly constrained analogs, (cyclo 5–9)-[Lys^5, Glu^9]-glucagon amide and (cyclo 17–21)-[Lys^{17}, Glu^{21}]-glucagon amide, were determined to have higher receptor binding affinities than glucagon, and higher helix contents in water–TFE mixtures.

Attempts to identify and characterize the amphiphilic α-helical region of calcitonin using lactam bridges have led to quite different and unexpected results that have significantly increased our understanding of calcitonin–receptor interactions.[34] Calcitonin is a 32-residue peptide hormone that is known primarily for its hypocalcemic effects, and is chemically synthesized for therapeutic use.[109,110] It is one of the first peptide hormones for which an amphiphilic α-helical structure was proposed for the bioactive conformation.[66,111] This helical structure was sug-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table III Pharmacological Activities of Calcitonin Analogs at the Human Calcitonin Receptor[a]

	$hCTR_{I1}$ Binding		$hCTR_{I1}$ Activity		
Peptide	$K_i \pm$ SE (n*M*)	Relative Affinity[b]	$ED_{50} \pm$ SE (n*M*)	Relative Potency[b]	K_i/ED_{50}[c]
Salmon calcitonin (sCT)	0.051 ± 0.009	100	0.115 ± 0.015	100	0.44
c(17–21)[Asp^{17}, Orn^{21}]–sCT	0.023 ± 0.006	220	0.091 ± 0.043	130	0.25
Human calcitonin (hCT)	42.5 ± 3.4	0.12	0.60 ± 0.10	19	71
c(17–21)[Asp^{17}, Lys^{21}]–hCT	195 ± 41	0.026	377 ± 148	0.031	0.52
[Nle^{8}, Phe^{12}, Asn^{17}, Nva^{21}, Tyr^{22}]-hCT	5.28 ± 1.75	0.97	0.27 ± 0.08	43	20
c(17–21)[Nle^{8}, Phe^{12}, Asp^{17}, Orn^{21}, Tyr^{22}]–hCT	0.56 ± 0.18	9.1	0.007 ± 0.002	1600	80

[a] Data taken from Ref. 34.
[b] Calculated relative to sCT = 100%.
[c] K_i/ED_{50} measures the agonist activity of the peptides compared to their affinities for the receptor.

gested for the entire central segment that lies between the disulfide loop formed by residues 1–7 at its N-terminal end, and a proline residue in position 23. Solution properties of the salmon peptide, sCT, and studies of receptor binding and activation by sCT analogs, have clearly established the stability of the helical conformation in this region of the salmon hormone,[110,112] and its role as a receptor binding element with a high affinity for the extracellular N-terminal domain of the human receptor.[113] However, as described earlier, our attempts to stabilize the corresponding helical structure in human calcitonin (hCT) by lactam-bridge constraints[33] led to the discovery that an (Asp^{17}, Lys^{21}) side-chain bridge actually *disfavored* helical structure in hCT and favored other ordered structures (Figure 8). Furthermore, this nonhelical analog exhibited higher potencies in some pharmacological assays *in vitro* and *in vivo,* including the hypocalcemic assay in mice.[33,63]

The ring-size requirements for the 17-to-21 side-chain bridge constraint in hCT were subsequently investigated by Kapurniotu and co-workers.[63] This study led to the identification of an analog, (*cyclo*-17-21)–[Asp^{17}, Orn^{21}]–hCT, that was over 500 times more potent than hCT itself in the hypocalcemic assay in mice, one hour after subcutaneous injection. The high potency of this analog, together with the smaller ring size of the constraining (Asp^{17}, Orn^{21}) bridge, which is incompatible with the formation of an α-helix,[29] led to the conclusion that the amphiphilic α helix proposed for the calcitonins must be terminated at or before residue 17. Circumstantial evidence based on NMR studies[73], and structure–activity data[63], suggest that the helix in hCT is terminated by a type I β-turn, centered around residues 18 and 19. This turn type may be stabilized by an H-bond to the native Asn^{17} in hCT, and by an H-bond to the Asp^{17} carbonyl in the potent (Asp^{17}, Orn^{21})-bridged analog. Recently, this conformational hypothesis has resulted in the development of a lactam-bridged hCT analog that is nearly 100-fold more potent than hCT in assays of receptor binding and signal transduction (adenylyl cyclase activation) at the cloned, insert-negative isoform of the human calcitonin receptor, $hCTR_{I1}$ (Table III).[34] These latest studies demonstrate that the (Asp^{17}, Orn^{21}) bridge constraint is fully compatible with the bioactive conformation of hCT: the increased potency of the lactam-bridged analog in the assay for adenylyl cyclase activation corresponds closely to that expected on the basis of its increased affinity for the receptors. As a result, the high ratio that is observed for the potency of hCT in the hCTR receptor activation assay compared to the affinity of hCT for these receptors (measured as K_iED_{50}) is retained in the potent (Asp^{17}, Orn^{21})-bridged hCT analog (Table III). The contrast with the (Asp^{17}, Lys^{21})-bridged hCT analog is striking, and suggests that the larger bridge results in a poorer fit of the folded (Asp^{17}, Lys^{21})-bridged hormone with the hCT binding site on $hCTR_{11}$, and with the requirements for receptor activation.

Salmon calcitonin binds to human calcitonin receptors with a high affinity compared to hCT. The binding element that is responsible for this high affinity has been mapped to the amphiphilic helical region, which interacts primarily with the N-terminal domain of the receptor (see Figure 16).[113] In comparison to hCT, however, the salmon peptide is also found to be much less effective in the signal transduction assay than would be expected on the basis of its high affinity for the hCTRs (Table III). Again, this suggests that sCT, which has only 50% sequence homology with hCT, is ineffective at meeting the conformational requirements for activation of the

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

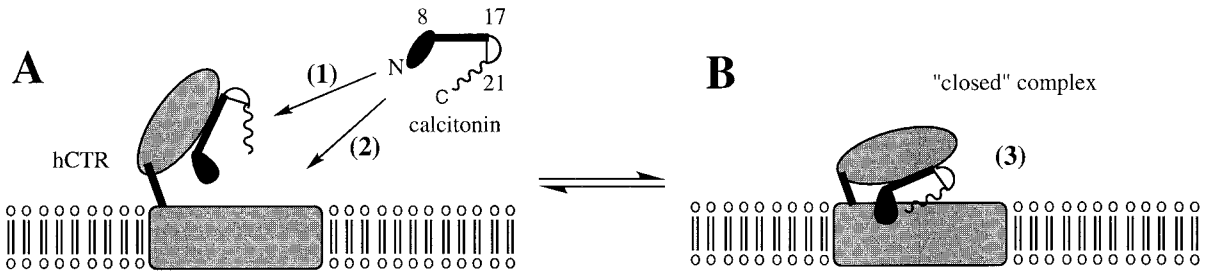


FIGURE 16 Model for receptor binding by human and salmon calcitonin. The structural domains of calcitonin are illustrated schematically as the N-terminal disulfide-bridged loop (N), the amphiphilic helix (8–17), the lactam-bridged turn (17–21), and the C-terminal domain (C). The N-terminal extracellular domain and the transmembrane domain of the human calcitonin receptor (hCTR) are also illustrated schematically. Hormone binding to the receptor is postulated to occur via binding of the amphiphilic helix to the N-terminal domain of the receptor (pathway **1**), or by direct location of binding sites in the transmembrane domain for the N- and C-terminal calcitonin structures (pathway **2**). Pathway **1** may be more prominent for sCT binding than for hCT binding. A tighter, "closed" complex is then postulated as a prerequisite for efficient receptor activation and signal transduction (**3**). Inefficient signal transduction by sCT analogs and by (*cyclo*-17-21)–[Asp^{17}, Lys^{21}]–hCT is proposed to result from the poor fit of these analogs in complex **3,** compared to that of hCT and (Asp^{17}, Orn^{21})-bridged hCT analogs (Table III and text).

hCTR. In contrast to hCT, the (Asp^{17}, Orn^{21}) bridge constraint had almost no effect on the potency of sCT at the hCTR (Table III).[34] This is explained by the hypothesized poor fit of the native sCT sequence to the requirements for activation of the hCTR, which the conformational constraint of this bridge cannot influence. The (Asp^{17}, Orn^{21}) bridge did result in a modest two-fold enhancement of the affinity of sCT for the hCTR, but this is much less than the affinity enhancement obtained for hCT. Again, a poor fit of the folded sCT peptide to the receptor would explain this result, since the bridged analog may not gain any advantage through the lowered entropy of its conformationally constrained unbound state if the bound state still retains a high level of disorder. A hypothetical model for receptor binding and activation by the salmon and human peptides and their (Asp^{17}, Orn^{21})-bridged analogs that is based on these structure–activity data[34,113] is presented in Figure 16.

The most significant progress toward a structural model for receptor binding and activation by an amphiphilic α-helical peptide hormone has, arguably, been achieved for PTH and PTHrP.[74,114–117] In these studies, PTH(1–34)–amide and PTHrP(1–34)–amide analogs have been synthesized with helix stabilizing (Glu^{i}, Lys^{i+4}) and (Lys^{i}, Glu^{i+4}) lactam bridges placed on the hydrophilic face of proposed amphiphilic α-helical regions and tested for receptor binding activity. The placement of a (Lys^{13}, Asp^{17}) bridge in PTHrP(7–34)-amide resulted in an antagonist potency five- to ten-fold higher than that of the linear peptide, suggesting a role for an N-terminal helical structure.[114] (Glu^{22}, Lys^{26}) and (Lys^{26}, Asp^{30}) lactam bridges incorporated into the C-terminal region PTH(1–31)–amide stabilize an amphiphilic helix that had previously been characterized for residues 17–29.[115] These analogs were also the most active hypotensive agents when tested *in vivo*. Similarly, dicyclic, (Lys^{18}, Asp^{22})- and (Lys^{26}, Asp^{30})-bridged analogs of PTH(1–31)–amide were potent agonists in adenylyl cyclase activation assays *in vivo*.[74] This information has then been combined with data from photoaffinity cross-linking experiments, in which photoactivatable benzophenone-modified PTH analogs have been used to map ligand–receptor interactions on the extracellular face of the receptor.[116,117] Together with NMR structural studies of the first extracellular loop of the receptor and receptor modeling, these approaches have provided distance constraints on the topology of the C-terminal helix interaction with the PTH receptor.[118,119] A model, based on the integrated information from these three approaches, positions the C-terminal amphiphilic α-helix in PTH between the first extracellular loop on the surface of the transmembrane domain of the receptor and the C-terminal helix of the extracellular N-terminal receptor domain. This model for the PTH–receptor complex is suggestive of a receptor binding role for the C-terminal amphiphilic α helix in PTH that is similar to that proposed in Figure 16 (following Ref. 113) for the corresponding helical structure in calcitonin. Indeed, it may be a general phenomenon that the N-terminal extracellular domains of the G-protein coupled receptors serve to bind the amphiphilic α-helical structures in this class

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of peptide hormones. This binding interaction may serve several functions, including (a) capture of the peptide ligand from solution or from the cell membrane surface, thereby effectively increasing the target size of the receptor for productive collisions by the peptide ligand; (b) enhancement of the affinity of the peptide hormone for its receptor; and (c) catalysis of the correct folding and orientation of the peptide ligand relative to its binding sites on the transmembrane domain of the receptor, which must be occupied for signal transduction. The increasing accuracy of the models being developed for the PTH–receptor complex, and the extension of these methods to studies of other peptide–receptor complexes will be needed to test these hypotheses.

CONCLUSIONS

Numerous convenient synthetic approaches are available for the solid-phase assembly of peptides that are conformationally constrained by multiple side-chain lactam bridges. The use of these bridges for stabilization of the α-helical conformation is well established, and conformational constraint of certain turn types and 3_{10}-helices has also been described. These conformational constraints have already found many applications to studies of peptide–protein recognition and protein folding. In particular, the lactam-bridge constraint of amphiphilic α-helical peptide hormones has aided considerably in the development of models for the interactions of these peptide ligands with their cell-surface receptors. The highly constrained α-helical structures that are formed when side-chain lactam bridges are placed in overlapping positions in the linear amino acid sequence are now synthetically accessible. These structures offer an important new opportunity to develop peptidomimetics for the α-helix and to explore helix propagation in simple peptide models of protein folding or of inaccessible α-helical biological systems.

Support from NIH grant R21 AI45316 is gratefully acknowledged.

REFERENCES

1. Hruby, V. J. Life Sci 1982, 31, 189–199.
2. Rizo, J.: Geirasch, L. M. Ann Rev Biochem 1992, 61, 387–418.
3. Ghadiri, M. R.; Granja, J. R.; Milligan, R. A.; McRee, D. A.; Khazanovich, N. Nature 1993, 366, 324–327.
4. Fernandez-Lopez, S.; Kim, H. S.; Choi, E. C.; Delgado, M.; Granja, J. R.; Khasanov, A.; Kraehenbuehl, K.; Long, G.; Weinberger, D. A.; Wilcoxen, K. M.; Ghadiri, M. R. Nature 2001, 412, 392–395.
5. Schiller, P. W.; DiMaio, J. Nature 1982, 297, 74–76.
6. Schiller, P. W.; Weltrowska, G.; Berezowska, I.; Nguyen, T. M.; Wilkes, B. C.; Lemieux, C.; Chung, N. N. Biopolymers 1999, 51, 411–425.
7. Mosberg, H. I. Biopolymers 1999, 51, 426–439.
8. Hruby, V. J. Acc Chem Res 2001, 34, 389–397.
9. Mapelli, C.; Kimura, H.; Stammer, C. H. Int J Pept Protein Res 1986, 28, 347–359.
10. Spolar, R. S.; Record, M. T., Jr. Science 1994, 263, 777–784.
11. Lin, S. H.; Konishi, Y.; Denton, M. E.; Scheraga, H. A. Biochemistry 1984, 23, 5504–5512.
12. Wetzel, R.; Perry, L. J.; Baase, W. A.; Becktel, W. J. Proc Natl Acad Sci USA 1988, 85, 401–405.
13. Kluger, R.; Stefano, V. D. J Org Chem 2000, 65, 214–219.
14. Camarero, J. A.; Fushman, D.; Sato, S.; Giriat, I.; Cowburn, D.; Raleigh, D. P.; Muir, T. W. J Mol Biol 2001, 308, 1045–1062.
15. Kumar, N. V.; Wemmer, D. E.; Kallenbach, N. R. Biophys Chem 1988, 31, 113–119.
16. Pease, J. H.; Wemmer, D. E. Biochemistry 1988, 27, 8491–8498.
17. Walker, C. S.; Steel, D.; Jacobsen, R. B.; Lirazan, M. B.; Cruz, L. J.; Hooper, D.; Shetty, R.; DelaCruz, R. C.; Nielsen, J. S.; Zhou, L. M.; Bandyopadhyay, P.; Craig, A. G.; Olivera, B. M. J Biol Chem 1999, 274, 30664–30671.
18. Camarero, J. A.; Cairo, J. J.; Giralt, E.; Andreu, D. J Peptide Sci 1995, 1, 241–250.
19. Barany, G.; Merrifield, R. B. In The Peptides; Gross, Meienhofer, eds.; Academic Press; New York, 1979, Vol 2, pp 1–284.
20. Stewart, J. M., Young, J. D. *Solid Phase Peptide Synthesis*; Pierce Chemical Co.: Rockford, IL, 1984; pp 71–95.
21. Tam, J. P.; Heath, W. F.; Merrifield, R. B. J Am Chem Soc 1983, 105, 6442–6455.
22. Tam, J. P.; Heath, W. F.; Merrifield, R. B. J Am Chem Soc 1986, 108, 5242–5251.
23. Fujii, N.; Otaka, A.; Ikemura, O.; Akaji, K.; Funakoshi, S.; Hayashi, Y.; Kuroda, Y.; Yajima, H. J Chem Soc Chem Commun 1987, 1987, 274–275.
24. Yajima, H.; Fujii, N.; Funakoshi, S.; Watanabe, T.; Murayama, E.; Otaka, A. Tetrahedron 1988, 44, 805–819.
25. Fields, G. B.; Noble, R. L. Int J Pept Protein Res 1990, 35, 161–214.
26. Schiller, P. W.; Nguyen, T. M.; Lemieux, C.; Maziak, L. A.; J Med Chem 1985, 28, 1766–1771.
27. Schiller, P. W.; Nguyen, T. M.-D.; Miller, J. Int J Peptide Protein Res 1985, 171–177.
28. Felix, A. M.; Heimer, E. P.; Wang, C. T.; Lambros, T. J.; Fournier, A.; Mowles, T. F.; Maines, S.; Camp-

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

bell, R. M.; Wegrzynski, B. B.; Toome, V.; Fry, D.; Madison, V. S. Int J Pept Protein Res 1988, 32, 441–454.
29. Campbell, R. M.; Bongers, J.; Felix, A. M. Peptide Sci 1995, 37, 67–88.
30. Story, S. C.; Aldrich, J. V. Int J Peptide Protein Res 1994, 43, 292–296.
31. Taylor, J. W.; Greenfield, N. J.; Wu, B.; Privalov, P. J Mol Biol 1999, 291, 965–976.
32. Yu, C.; Taylor, J. W. Bioorg Med Chem 1999, 7, 161–175.
33. Kapurniotu, A.; Taylor, J. W. J Med Chem 1995, 38, 836–847.
34. Taylor, J. W.; Jin, Q. K.; Sbacchi, M.; Wang, L.; Belfiore, P.; Garnier, M.; Kazantzis, A.; Kapurniotu, A.; Zaratin, P. F.; Scheideler, M. A. J Med Chem 2002, 45, 1108–1121.
35. Cavallaro, V.; Thompson, P.; Hearn, M. J Peptide Sci 1998, 4, 335–343.
36. Zhang, W.; Taylor, J. W. Tetrahedron Lett 1996, 37, 2173–2176.
37. Osapay, G.; Taylor, J. W. J Am Chem Soc 1990, 112, 6046–6051.
38. Osapay, G.; Taylor, J. W. J Am Chem Soc 1992, 114, 6966–6973.
39. Bouvier, M.; Taylor, J. W. J Med Chem 1992, 35, 1145–1155.
40. Kapurniotu, A.; Taylor, J. W. Tetrahedron Lett 1993, 34, 7031–7034.
41. Felix, A. M.; Wang, C. T.; Heimer, E. P.; Campbell, R. M.; Madison, V. S.; Fry, D.; Toome, V.; Downs, T. R.; Frohman, L. A. In Peptides: Chemistry Structure and Biology, Rivier, J., Marshall, G. R. Eds.; ESCOM Publishers BV: Leiden, 1990, pp 226–227.
42. Guibé, F. Tetrahedron 1998, 54, 2967–3042.
43. Grieco, P.; Gitu, P. M.; Hruby, V. J. J Peptide Res 2001, 57, 250–256.
44. Ahn, J. M.; Gitu, P. M.; Medieros, M.; Swift, J. R.; Trivedi, D.; Hruby, V. J. J Med Chem 2001, 44, 3109–3116.
45. Zhang, M.; Wu, B.; Baum, J.; Taylor, J. W. J Peptide Res 2000, 55, 398–406.
46. Kunz, H.; Kneip, M. Angew Chem Int Ed Engl 1984, 23, 716–717.
47. Kessler, H.; Becker, G.; Kogler, H.; Wolff, M. Tetrahedron Lett 1984, 25, 3971–3974.
48. Yu, C.; Taylor, J. W. Tetrahedron Lett 1996, 37, 1731–1734.
49. Kaiser, E.; Colescott, R. L.; Bossinger, C. D.; Cook, P. I. Anal Biochem, 1970, 34, 595–598.
50. Rink, H. Tetrahedron Lett 1987, 28, 3787–3790.
51. Taylor, J. W.; Reddy, P.; Patel, K.; Dineen, T.; Naqvi, S. *In* Peptides: The Wave of the Future; *Proceedings of the 2nd International/17th American Peptide Symposium*; Lebl, M., Houghten, R. A., Eds.; Kluwer Academic Publishers: Norwell, MA, 2002; pp 67–68.
52. Isied, S. S.; Vassilian, A.; Lyon, J. M. J Am Chem Soc 1982, 104, 3910–3916.
53. Tian, Y.; Ramesh, C. V.; Ma, X.; Patel, T.; Tiscione, M.; Cenizal, T.; Taylor, J. W.; Arnold, G. F.; Arnold, E. *In* Peptides: The Wave of the Future; *Proceedings of the 2nd International/17th American Peptide Symposium*; Lebl, M., Houghten, R. A., Eds.; Kluwer Academic Publishers: Norwell, MA, 2002; 1016–1017.
54. Tian, Y.; Ramesh, C. V.; Ma, X.; Naqvi, S.; Patel, T.; Cenizal, T.; Tiscione, M.; Diaz, K.; Crea, T.; Arnold, E.; Arnold, G. F.; Taylor, J. W. J Peptide Res, 2002, 59, 264–276.
55. Ghadiri, M. R.; Choi, C. J Am Chem Soc 1990, 112, 1630–1632.
56. Ghadiri, M. R.; Fernholz, H. J Am Chem Soc 1990, 112, 9633–9635.
57. Kohn, W. D.; Kay, C. M.; Sykes, B. D.; Hodges, R. S. J Am Chem Soc 1998, 120, 1124–1132.
58. Kelso, M. J.; Hoang, H. N.; Appleton, T. G.; Fairlie, D. P. J Am Chem Soc 2000, 122, 10488–10489.
59. Cabezas, E.; Satterthwait, A. C. J Am Chem Soc 1999, 121, 3862–3875.
60. Fry, D. C.; Madison, V. S.; Greeley, D. N.; Felix, A. M.; Heimer, E. P.; Frohman, L.; Campbell, R. M.; Mowles, T. F.; Toome, V.; Wegrzynski, B. B. Biopolymers 1992, 32, 649–666.
61. Houston, M. E., Jr.; Gannon, C. L.; Kay, C. M.; Hodges, R. S. J. Peptide Sci 1995, 1, 274–282.
62. Taylor, J. W.; Guo, G.; Wu, B.; Yu, C.; Zhang, M.; Zhang, W. In *Peptidomimetics & Small Molecule Design*; Hori, W., ed.; IBC library Series, Publication #984; International Business Communications: Southborough, MA, 1997; pp 87–120.
63. Kapurniotu, A.; Kayed, R.; Taylor, J. W.; Voelter, W. Eur J Biochem 1999, 265, 606–618.
64. Bracken, C.; Gulyás, J.; Taylor, J. W.; Baum, J. J Am Chem Soc 1994, 116, 6431–6432.
65. O'Neil, K. T.; DeGrado, W. F. Science 1991, 250, 646–651.
66. Kaiser, E. T.; Kezdy, F. J. Science 1984, 223, 249–255.
67. Wu, B. Y.; Gaffney, B. L.; Jones, R. A.; Taylor, J. W. In *Proceedings of the 14th American Peptide Symposium, Peptides: Chemistry, Structure and Biology*; Kaumaya, P. T. P., Hodges, R. S., Eds.; Mayflower Scientific, England, 1996; pp 265–266.
68. Zhang, M.; Wu, B.; Zhao, H.; Taylor, J. W. J Peptide Sci 2002, 8, 125–136.
69. Houston, M. E., Jr.; Wallace, A.; Bianchi, E.; Pessi, A.; Hodges, R. S. J Mol Biol 1996, 262, 270–282.
70. Taylor, J. W,; Guo, G.; Yu, C. In *Proceedings of the 15th American Peptide Symp*; (Tam, J. P., Kaumaya, P. T. P., Eds.); Kluwer Academic Publishers: Boston, 1998; pp 364–365.
71. Evans, C. J.; Keith, D. E.; Morrison, H.; Magendzo, K.; Edwards, R. H. Science 1992, 258, 1952–1955.
72. Sueki, M.; Lee, S.; Powers, S. P.; Denton, J. B.; Konishi, Y.; Scheraga, H. A. Macromolecules 1984, 30, 121–132.
73. Motta, A.; Temussi, P. A.; Wünsch, E.; Bovermann, G. Biochemistry 1991, 30, 2364–2371.

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

74. Condon, S.; Morize, I.; Darnbrough, S.; Burns, C. J.; Miller, B. E.; Uhl, J.; Burke, K.; Jariwala, N.; Locke, K.; Krolokowski, P. H.; Kumar, N. V.; Labaudiniere, R. F. J Am Chem Soc 2000, 122, 3007–3014.
75. Jackson, D. Y.; King, D. S.; Chmielewski, J.; Singh, S.; Schultz, P. G. J Am Chem Soc 1991, 113, 9391–9392.
76. Phelan, J. C.; Skelton, N. J.; Braisted, A. C.; McDowell, R. S. J Am Chem Soc 1997, 119, 455–460.
77. Fairman, R.; Anthony-Cahill, S. J.; DeGrado, W. F.; J Am Chem Soc 1992, 114, 5458–5468.
78. Zhang, M.; Yu, C.; Baum, J.; Taylor, J. W. In *Proceedings of the 16th American Peptide Symposium*; Fields, G. B., Tam, J. P., Barany, G., Eds., Kluwer Academic Publishers; Boston, 2000; pp 283–284.
79. Luo, P.; Braddock, D. T.; Subramanian, R. M.; Meredith, S. C.; Lynn, D. G. Biochemistry 1994, 33, 12367–12377.
80. Luo, P.; Baldwin, R. L. Biochemistry, 1997, 36, 8413–8421.
81. Schievano, E.; Bisello, A.; Chorev, M.; Bisol, A.; Mammi, S.; Peggion, E. J Am Chem Soc 2001, 123, 2743–2751.
82. Gulyas, J.; Rivier, C.; Perrin, M.; Koerber, S. C.; Sutton, S.; Corrigan, A.; Lahrichi, S. L.; Craig, A. G.; Vale, W.; Rivier, J. Proc Natl Acad Sci USA 1995, 92, 10575–10579.
83. Marepalli, H. R.; Antohli, O.; Becker, J. M.; Naider, F. J Am Chem Soc 1996, 118, 6531–6539.
84. Antohi, O.; Marepalli, H. R.; Yang, W.; Becker, J. M.; Naider, F. Biopolymers 1998, 45, 21–34.
85. Kemp, D. S.; Allen, T. J.; Oslick, S. L.; Boyd, J. D. J Am Chem Soc 1996, 118, 4240–4248.
86. Kemp, D. S.; Oslick, S. L.; Allen, T. J. J Am Chem Soc 1996, 118, 4249–4255.
87. Cabezas, E.; Chiang, L.-C.; Satterthwait, A. C. In Peptides. Chemistry, Structure and Biology. Proceeding of the 13th American Peptide Society. (Hodges, R., Smith, J., A. Eds.); ESCOM: Leiden, 1994; pp 287–289.
88. Chitnumsub, P.; Fiori, W. R.; Lashuel, H. A.; Diaz, H.; Kelly, J. W. Bioorg Med Chem 1999, 7, 39–59.
89. Tsai, J. H.; Waldman, A. S.; Nowick, J. S. Bioorg Med Chem 1999, 7, 29–38.
90. Phillips, S. T.; Rezac, M.; Abel, U.; Kossenjans, M.; Bartlett, P. J Am Chem Soc 2002, 124, 58–66.
91. Paterlini, G.; Portoghese, P. S.; Ferguson, D. M. J Med Chem 1997, 40, 3254–3262.
92. Baldwin, J. M. EMBO J 1993, 12, 1693–1703.
93. Pogozheva, I. D.; Lomize, E. L.; Mosberg, H. I. Biophys J 1998, 75, 612–634.
94. Chan, D. C.; Kim, P. S. Cell 1998, 93, 681–684.
95. Chan, D. C.; Fass, D.; Berger, J. M.; Kim, P. S. Cell 1997, 89, 263–273.
96. Weissenhorn, W.; Dessen, A.; Harrison, S. C.; Skehel, J. J.; Wiley, D. C. Nature 1997, 387, 426–430.
97. Muster, T.; Steindl, F.; Purtscher, M.; Trkola, A.; Klima, A.; Himmler, G.; Ruker, F.; Katinger, H. J Virol 1993, 67, 6642–6647.
98. Pai, E. F.; Klein, M. H.; Chong, P.; Pedyczak, A. World Intellectual Property Organization (www.wipo.org) patent WO-00/61618, 2000
99. Yu, Y. B.; Lavigne, P.; Kay, C. M.; Hodges, R. S.; Privalov, P. L. J Phys Chem 1999, 103, 2270–2278.
100. Scholtz, J. M.; Marqusee, S.; Baldwin, R. L.; York, E. J.; Stewart, J. M.; Santoro, M.; Bolen, D. W. Proc Natl Acad Sci USA 1991, 88, 2854–2858.
101. Shalongo, W.; Dugad, L.; Stellwagen, E. J Am Chem Soc 1994, 116, 2500–2507.
102. Loladze, V. V.; Ermolenko, D. N.; Makhatadze, G. I. Protein Sci 2001, 1343–1352.
103. Williams, L.; Kather, K.; Kemp, D. S. J Am Chem Soc 1998, 120, 11033–11043.
104. Wild, C. T.; Shugars, D. C.; Greenwell, T. K.; McDanal, C. B.; Matthews, T. J. Proc Natl Acad Sci USA, 1994, 91, 9770–9774.
105. Wild, C.; Greenwell, T.; Shugars, D.; Rimsky-Clarke, L.; Matthews, T. AIDS Res Hum Retroviruses 1995, 11, 323–325.
106. Judice, J. K.; Tom, J. Y.; Huang, W.; Wrin, T.; Vennari, J.; Petropoulos, C. J.; McDowell, R. S. Proc Natl Acad Sci USA 1997, 94, 13426–13430.
107. Taylor, J. W.; Kaiser, E. T. Pharmacol Rev 1986, 38, 291–319.
108. Taylor, J. W. In *The Amphipathic Helix*; Epand, R. M., Ed.; CRC Press: Boca Raton, FL, 1993; pp Chap 11, 285–311.
109. Azria, M. *The Calcitonins: Physiology and Pharmacology*; Karger: Basel, 1989; pp 133–143.
110. Sexton, P. M.; Findlay, D. M.; Martin, T. J. Curr Med Chem 1999, 6, 1067–1093.
111. Moe, G. R.; Kaiser, E. T. J Am Chem Soc 1983, 105, 4100–4102.
112. Epand, R. M.; Epand, R. F.; Orlowski, R. C. Int J Pept Protein Res 1985, 25, 105–111.
113. Stroop, S. D.; Nakamuta, H.; Kuestner, R. E.; Moore, E. E.; Epand; R. M. Endocrinology 1996, 137, 4752–4756.
114. Chorev, M.; Roubini, E.; McKee, R. L.; Gibbons, S. W.; Goldman, M. E.; Caulfield, M. P.; Rosenblatt, M. Biochemistry 1991, 30, 5968–5974.
115. Barbier, J.-R.; Neugebauer, W.; Morley, P.; Ross, V.; Soska, M.; Whitfield, J. F.; Willick, G. J Med Chem 1997, 40, 1379–1380.
116. Zhou, A. T.; Bessalle, R.; Bisello, A.; Nakamoto, C.; Rosenblatt, M.; Suva, L. J.; Chorev, M. Proc Natl Acad Sci USA 1997, 94, 3644–3649.
117. Greenberg, Z.; Bisello, A.; Mierke, D. F.; Rosenblatt, M.; Chorev, M. Biochemistry 2000, 39, 8142–8152.
118. Piserchio, A.; Bisello, A.; Rosenblatt, M.; Chorev, M.; Mierke, D. F. Biochemistry 2000, 39, 8153–8160.
119. Monticelli, L.; Mammi, S.; Peggion, E.; Ferrari, S.; Rosenblatt, M.; Chorev, M.; Bisello, A.; Mierke, D. F. *In* Peptides: The Wave of the Future; *Proceedings of the 2nd International/17th American Peptide Symposium*; Lebl, M., Houghten, R. A., Eds.; Kluwer Academic Publishers; Norwell, MA, 2002; pp 424–425.

10970282a, 2002, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/bip.10203 by Ocean University Of China, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License